Siddaganga Institute of Technology, Tumakuru

(An Autonomous institution affiliated to Visvesvaraya Technological University, Belagavi, Approved by AICTE, New Delhi, Accredited by NAAC and ISO 9001:2015 certified)

Mini Project Report on

Mobile app for checking seats in buses

submitted

in the partial fulfillment of the requirements VI semester

Bachelor of Engineering

In

Computer Science and Engineering by

Sahana A S 1SI18CS097

Sampath H S 1SI18CS098

Shifa Ilman 1SI18CS103

under the guidance of

Prof. Nayana S M. Tech

Assistant Professor

Department of Computer Science & Engineering

(Program Accredited by NBA)

Siddaganga Institute of Technology

B.H Road, Tumakuru-572 103, Karnataka, India.

Web: www.sit.ac.in

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Certificate

This is to certify that the mini project entitled "Mobile app for checking seats in buses" is a bonafide work carried out by **Sahana A S (1SI18CS097), Sampath H S (1SI18CS098), Shifa Ilman (1SI18CS103)** of VI semester **Computer Science and Engineering**, **SIDDAGANGA INSTITUTE OF TECHNOLOGY** during the academic year 2020-2021.

Signature of the Guide

Prof. Nayana S M. Tech

Assistant Professor

Signature of the Convener

Prof. Bhaskar G M. Tech

Assistant Professor

Signature of the HOD

Dr. A S Poornima

Prof. and Head, Dept. of CSE

Name of the Examiners: **Signature with Date**

1. Prof.
2. Prof.

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With reverential pranams, we express our sincere gratitude and salutations to His Holiness **Dr. Sree Sree Shivakumara Swamigalu,** Founder President of Sree Siddaganga Education Society and **Sree Sree Siddalinga Swamigalu**, President of Sree Siddaganga Education Society for their unlimited blessings.

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-Sahana A S(1SI18CS097)

-Sampath H S(1SI18CS098)

-Shifa Ilman(1SI18CS103)

ABSTRACT

Since the rise of information technology, manual system has turned out to be exercise in futility and vitality particularly in the created nations. The fundamental purpose behind not building up a robotized system is expected the absence of learning and inventiveness in information technology field. The essential point to carry out this venture is to make and plan basically by building a Bus booking system of Android Application for the organizations, keeping in mind the end goal to exchange all their manual undertakings or routine operations into computerized system, which will enable the organization to serve its clients up to their ideal fulfillment. It is a typical thing that information technology has changed numerous manual operations to mechanized and individuals are getting a charge out of it much better that the manual system.

The proposed system will serve and remain as a worldwide system communication arrangement of the organization with clients. The system allows users to book ticket on their fingertips without getting stress or going to the counter to purchase a ticket. Once you register by filling in all the necessary information you become a user by having all the access to use the system. The system has so many functionality such as - the bottom shows on application by knowing the routes, knowing the price and discount also displaying when booking, payment also is been provided all in the system.

The system contains admin part which takes all the responsibility for updating most the information need for the benefit of the customers and also allow users to know all the information where each bus is going to and also by knowing the destination. The system also shows on how to get to each stop and give users the fares for each route and also is trickily based on the location that is been provided for a user's.

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CHAPTER-1

INTRODUCTION

The “Bus Ticket Booking System” has been developed to overcome the problems prevailing in the practicing manual system. This software is supported to eliminate and, in some cases, reduce the hardships faced by this existing system. Moreover, this system is designed for the particular need of the company to carry out operation in a smooth and effective manner.

The application is reduced as much as possible to avoid errors while entering the data. It also provides error message while entering invalid data. No formal knowledge is needed for the user to use this system. Thus, by this all it proves it is user-friendly. Bus Ticket Booking System, as described above, can lead to error free, secure, reliable and fast management system. It can assist the assist the user to concentrate on their other activities rather to concentrate on the record keeping. Thus, it will help organization in better utilization of resources.

Most monetary transactions on the world have a tendency to be electronic, in light of the fact that it gives effectiveness, security, precision and unwavering quality.in Sudan staffs at the ticket counter are using the manual system to sell tickets and manage the bus seat booking. Travelling companies spends a substantial cost to manage the reservation process.

Online bus reservation system allows customers to reserve seats, buy and pay tickets online the organization can give reservation administrations and data to their clients without the limitation of office hours or manpower. Not just does it let clients book trips around the clock from any location with an internet connection. yet it is additionally designed for use by the company to internally manage their business processes; minimizing human errors and overcoming difficulties and problems that arose in the previous system.

Every organization, whether big or small, has challenges to overcomes and managing the information of Tickets, Customer, Agent, Seats. Every bus Ticket Booking System has different bus needs therefore we design exclusive employee management systems that are adapted to your managerial requirements. This is designed to assist in strategic planning and will help you ensure that your organization is equipped with the right level of information

and details for your future goals. Also, for those Busy executive who are always on the go, our systems come with remote access features, which will allow you to manage your workforce anytime, always. This system will ultimately allow you to better manage resources.

Objectives:

- The main objective of the project is to check number of seats in buses and to manage the details of bus and the seats.
- It manages all the information about vacancy of seats in busses.
- The project is totally built at administration end and thus the administration is guaranteed the access.
- The purpose of project is to built n application program to reduce the manual work for managing the bus, ticket, checking seats in buses.
- It tracks all the information about checking seats in buses.
- Providing web-based seat checking system where customer can check the number of available seats and boot them through online system without a need to queue up at the counter to purchase a bus ticket.

Motivation:

Craze for travel is not new, but now the era is changed. People with growing trends are option lifestyle online. In this regard, booking transport facilities online is not a wonder. So, to ease this venture, we have developed an android app to facilitate booking to destination and ease the struggle to travel. The features in the app help to reach destination with more precise and affordable pricing with consumer-friendly experience.

In this app, we not only provide customer facility for advanced booking of vacant seats but also cancellation of reservation with proper money refunds. We provide several buses for customer to provide them comfortable journey.

Online mode of payment is available, money is refunded immediately. We provide a hand in option to print ticket also. In case, after a second thought if the customer wants to update the destination or bus chosen, facility for this also will be provided. Payment updations are also considered.

CHAPTER-2

LITERATURE SURVEY

[1] "A survey on Mobile Application Development using Android OS" By Sujithra Ma, Padmavathi G-. Satyanarayana

Smart phone has become the most typical and Popular mode of communication that combines the functionalities of both mobile devices and the personal computer. Smart phone provides many features such processing, communication, data storage with the services like browser, portable media player, video call, GPS, WI-FI etc. Worldwide penetration of smart phones and PDAs in recent years still having a question for their mobile platform standards. The increasing importance of smart phone has triggered intense competition among technology giants like Symbian, Microsoft, Apple, Nokia, Blackberry and Android. In a bid to capture the bigger market share for mobile platform. Android, a mobile operating system based on the Linus platform developed by Google is an Open Handset Alliamce. This paper discusses the various aspects of Android architecture, Programming framework, features and also a comparation analysis with other mobile platform standard.

[2] "Software Engineering Issues for Mobile Application Development" By Anthony I. Wasserman

This paper provides an overview of important software engineering research issues related to be development of applications that run on mobile devices. Among the topics are development process, tools, user interface design, application portability, quality, and security.

[3] "Developing successful mobile application" By Harri Oinas Kukkonen

This paper discusses issues related to developing Successful mobile applications both business and in terms of user satisfaction. It shows that despite the many limitations mobile devices have, it is worthwhile considering developing them. Mobile commerce may become the key driving force for developing mobile applications, just as electronic commerce

catalyzed the development of Web applications. This piper puts special emphasis on goal driven applications and suggested seven key principles for developing highly goal-driven mobile applications. Finally, the importance of mobile usability evaluation is emphasized.

[4] "Adopting an Agile Approach for the Development of Mobile Applications." By Swati V. Chande

With the increasing popularity and demand for mobile applications, there has been a significant increase in the number of mobile application development project. Highly volatile requirements of mobile applications require adaptive software development methods. The Agile approach is seen as a natural for mobile application and there is a need to explore various Agile methodologies for the developments of mobile applications. This proper evaluates adopting various agile approaches improves the development of mobile applications and if they can be used in order to provide more tailoc made process improvements within an organization. A survey related mobile application development process improvement was developed. The use of various Agile approaches for success in mobile application development were evaluated by determining the significance of the most used Agile engineering paradigms such as XP, Scrum, and Lean. The finding of the way study shows that this Agile methods have the potential so help deliver enhanced speed and quality for mobile application development.

The database searched for literature review in this study is Scopus (https://www.scopus.com) which is currently the greatest and the most significant database for academic journals. Scopus records papers into various classifications, for this study we have considered sections (I) Computer Science (ii) Engineering & crossdisciplinary research, and this was another purpose behind choosing Scopus for this study.

As the range of our point is spoken to by these two classifications, the utilization of Scopus has been a characteristic decision. There are other digital databases like Web of Science, DBLP and IEEE Xplore however Scopus contains more academic journals than these databases. The hunt was performed autonomously on Scopus utilizing 'or' administrator for every watchword and from there on the query items of these two ideas was blended utilizing an 'and' administrator in Scopus. To reproduce this inquiry on Scopus.com, this punctuation can be reordered in the propelled hunt area of that digital database. The query item will change in the number of archives as this database is effectively refreshed.

The information for this study has been taken from the primary conceivable date of indexation of a journal paper to the date of inquiry performed on the Scopus database (Jan 2, 2019). Hence, this information is a genuine impression of the data that was exhibited on Jan 2, 2019.This brought about 29580 hits. The second stage, comprised of a look for HSC catchphrases inside the after-effects of the principal arrange, bringing about 3124 hits. In the third stage, we have seen the convergence of the information from stage one and stage two yielded in 2207 papers. In stage four, we have considered just journal articles with (322 papers) with numerous filtrations for the literature review and disposed of meeting procedures. Finally, only 26 papers were observed to be fit and relevant in the specific area of research.

Little research, however, has been conducted to define the specific elements that constitute effective website design. One of the key design measures is usability (International Standardization Organization, 1998). The International Standardized Organization (ISO) defines usability as the extent to which users can achieve desired tasks (e.g., access desired information or place a purchase) with effectiveness (completeness and accuracy of the task), efficiency (time spent on the task), and satisfaction (user experience) within a system. However, there is currently no consensus on how to properly operationalize and assess website usability (Lee & Kozar, 2012).

For example, Nielson associate's usability with learnability, efficiency, memorability, errors, and satisfaction (Nielsen, 2012). Yet, Palmer (2002) postulates that usability is determined by download time, navigation, content, interactivity, and responsiveness. Similar to usability, many other key design elements, such as scannability, readability, and visual aesthetics, have not yet been clearly defined (Bevan, 1997; Brady & Phillips, 2003; Kim, Lee, Han, & Lee, 2002), and there are no clear guidelines that individuals can follow when designing websites to increase engagement.

CHAPTER-3

REQUIREMENT SPECIFICATION

Tools & Technologies

Hardware Requirements

Processor	Intel core due 2.0 GHz or More
RAM	8 GB or More
Hard disk	120 GB or More
Monitor	15”CRT, or LCD monitor
Keyboard	Normal or Multimedia
mouse	Compatible mouse

Software Requirements

Front End	Android Studio, Flutter, Dart, Android Emulator with min SDK 21
Back End	Firebase, Razoypay (Payment), Wiredash
Operating System	Windows, Android
Browser	Internet Explorer or Google Chrome

Firebase:

Google Firebase is a Google-backed application development software that enables developers to develop iOS, Android and Web apps. Firebase provides tools for tracking analytics, reporting and fixing app crashes, creating marketing and product experiment.

Firebase offers a number of services, including:

- Analytics – Google Analytics for Firebase offers free, unlimited reporting on as many as 500 separate events. Analytics presents data about user behavior in iOS

and Android apps, enabling better decision-making about improving performance and app marketing.

- Authentication – Firebase Authentication makes it easy for developers to build secure authentication systems and enhances the sign-in and onboarding experience for users. This feature offers a complete identity solution, supporting email and password accounts, phone auth, as well as Google, Facebook, GitHub, Twitter login and more.
- Cloud messaging – Firebase Cloud Messaging (FCM) is a cross-platform messaging tool that lets companies reliably receive and deliver messages on iOS, Android and the web at no cost.
- Realtime database – the Firebase Realtime Database is a cloud-hosted NoSQL database that enables data to be stored and synced between users in real time. The data is synced across all clients in real time and is still available when an app goes offline.
- Crashlytics – Firebase Crashlytics is a real-time crash reporter that helps developers track, prioritize and fix stability issues that reduce the quality of their apps. With crashlytics, developers spend less time organizing and troubleshooting crashes and more time building features for their apps.
- Performance – Firebase Performance Monitoring service gives developers insight into the performance characteristics of their iOS and Android apps to help them determine where and when the performance of their apps can be improved.
- Test lab – Firebase Test Lab is a cloud-based app-testing infrastructure. With one operation, developers can test their iOS or Android apps across a variety of devices and device configurations. They can see the results, including videos, screenshots and logs, in the Firebase console.

Advantages:

1. Reliable & Extensive Databases

Firebase works under the flag of Google, and that's why it provides powerful databases for web and mobile application development. We are going to discuss major Firebase databases with their advantages below:

Realtime Database

No doubt, Realtime database was the first product that didn't lose its charm till now. Firebase Realtime database allows applications to approach cross-platform data within realtime after joining NoSQL cloud-storage. This Realtime database also enables you to work without internet connectivity. Even data is still being cached in the memory of your device when you are offline and start synchronizing after the internet connectivity.

Cloud Firestore

Another advantage of Firebase is its Cloud Firestore. This NoSQL database makes it easy for programmers to transfer and store data for front and backend development. This cloud database is also known for its Realtime updates, flexible data models, offline support, and quick data inquiries.

2. Fast & Safe Hosting

Another appealing advantage of Firebase is its secure & fast hosting services. Firebase hosting supports all content types, including web applications, dynamic and static content. Moreover, either you want to host your Express.js microservices, HTML, CSS, or APIs, the hosting support of Firebase is always there. It means Firebase hosts a diverse variety of content.

3. Provides A Free Start to Newbies

If you are a newbie programmer and willing to know the core advantage of Firebase, then you consider its "free start" feature. Yes, Firebase allows coders to develop their applications without investing a single penny. With the free services of Firebase, it is convenient for beginners to understand how their app works in a real environment.

4. Google Analytics

Although Google Analytics is a free tool provided by Google but its integration becomes seamless when you work with Firebase. Google Analytics is compatible with Ios,

Android, Web, C++ and Unity setups. This free analytic solution reports the developers how users are behaving towards their mobile and web applications. Firebase Analytics is also beneficial to improve retention and engagement rates of users for your application.

5. Firebase Cloud Messaging for Cross-Platform

Are you looking for a cross-platform messaging solution to send your notifications and messages at no cost? If yes, then Firebase Cloud Messaging (FCM) is the smartest solution for you. The key capabilities of FCM consist of sending simple notification messages, sending notifications from client apps, and versatile message targeting.

Because of the following three message sharing options, versatile message targeting is very popular among the users:

- Allows single device message distribution.
- Permits message distribution to a group of devices.
- Sends messages to devices which subscribe to the related topics.

Flutter:

Flutter is an open-source UI software development kit created by Google. It is used to develop cross platform applications for Android, Ios, Linux, Mac, Windows, Google Fuchsia, and the web from a single codebase.

The first version of Flutter was known as codename “Sky” and ran on the Android operating system. It was unveiled at the 2015 Dart developer summit with the stated intent of being able to render consistently at 120 frames per second. During the keynote of Google Developer Days in Shanghai, Google announced Flutter Release Preview 2, which is the last big release before Flutter 1.0. On December 4, 2018, Flutter 1.0 was released at the Flutter Live event, denoting the first “stable” version of the Framework. On December 11, 2019, Flutter 1.12 was released at the Flutter Interactive event.

On May 6, 2020, the Dart software development kit (SDK) in version 2.8 and the Flutter in version 1.17.0 were released, where support was added to the Metal API, improving performance on Ios devices (approximately 50%), new Material widgets, and new network tracking.

On March 3, 2021, Google released Flutter 2 during an online Flutter Engage event. This major update brought official support for web-based applications with new CanvasKit renderer and web specific widgets, early-access desktop application support for Windows, macOS, and Linux and improved Add-to-App APIs. This release included sound null-safety, which caused many breaking changes and issues with many external packages, but the Flutter team included instructions to mitigate these changes as well.

Dart Language:

Flutter apps are written in the Dart language and make use of many of the language"s more advanced features.

On Windows, macOS, and Linux Flutter runs in the Dart virtual machine, which features a just-in-time execution engine. While writing and debugging an app, Flutter uses Just in Time compilation, allowing for "hot reload", with which modifications to source files can be injected into a running application. Flutter extends this with support for stateful hot reload, where in most cases changes to source code are reflected immediately in the running app without requiring a restart or any loss of state.

For better performance, release versions of Flutter apps targeting Android and Ios are compiled with ahead-of-time (AOT) compilation.

Dart is an open-source, general-purpose, object-oriented programming language with C-style syntax developed by Google in 2011. The purpose of Dart programming is to create a frontend user interfaces for the web and mobile apps. It is under active development, compiled to native machine code for building mobile apps, inspired by other programming languages such as Java, JavaScript, C#, and is Strongly Typed. Since Dart is a compiled language so you cannot execute your code directly; instead, the compiler parses it and transfer it into machine code.

It supports most of the common concepts of programming languages like classes, interfaces, functions, unlike other programming languages. Dart language does not support arrays directly. It supports collection, which is used to replicate the data structure such as arrays, generics, and optional typing.

CHAPTER-4

DESIGN AND IMPLEMENTATION

ANDROID STUDIO:

Android Studio is the official Integrated Development Environment (IDE) for Android app development, based on IntelliJ IDEA.

To support application development within the Android operating system, Android Studio uses a Gradle-based build system, emulator, code templates, and Github integration. Every project in Android Studio has one or more modalities with source code and resource files. These modalities include Android app modules, Library modules, and Google App Engine modules.

Android Studio uses an Instant Push feature to push code and resource changes to a running application. A code editor assists the developer with writing code and offering code completion, refraction, and analysis. Applications built in Android Studio are then compiled into the APK format for submission to the Google Play Store.

On top of IntelliJ's powerful code editor and developer tools, Android Studio offers even more features that enhance your productivity when building Android apps, such as:

- A flexible Gradle-based build system.
- A fast and feature-rich emulator.
- A unified environment where you can develop for all Android devices.
- Apply Changes to push code and resource changes to your running app without restarting your app.
- Code templates and GitHub integration to help you build common app features and import sample code.
- Extensive testing tools and frameworks.
- Lint tools to catch performance, usability, version compatibility, and other problems.

- C++ and NDK support.
- Built-in support for Google Cloud Platform, making it easy to integrate Google Cloud Messaging and App Engine.

ANDROID EMULATOR:

The Android Emulator simulates Android devices on your computer so that you can test your application on a variety of devices and Android API levels without needing to have each physical device.

The emulator provides almost all the capabilities of a real Android device. You can simulate incoming phone calls and text messages, specify the location of the device, simulate different network speeds, simulate rotation and other hardware sensors, access the Google Play Store, and much more.

Testing your app on the emulator is in some ways faster and easier than doing so on a physical device. For example, you can transfer data faster to the emulator than to a device connected over USB.

Requirements and recommendations:

The Android Emulator has additional requirements beyond the basic system requirements for Android Studio, which are described below:

- SDK Tools 26.1.1 or higher.
- 64-bit processor.
- Windows: CPU with UG (unrestricted guest) support.
- HAXM 6.2.1 or later (HAXM 7.2.0 or later recommended)

The use of hardware acceleration has additional requirements on Windows and Linux:

- Intel processor on Windows or Linux: Intel processor with support for Intel VT-x, Intel EM64T (Intel 64), and Execute Disable (XD) Bit functionality
- AMD processor on Linux: AMD processor with support for AMD Virtualization

(AMD-V) and Supplemental Streaming SIMD Extensions 3 (SSSE3).

- AMD processor on Windows: Android Studio 3.2 or higher and Windows 10 April 2018 release or higher for Windows Hypervisor Platform (WHPX) functionality

To work with Android 8.1 (API level 27) and higher system images, an attached webcam must have the capability to capture 720p frames.

Deprecation for 32-bit Windows systems:

The Android Emulator was deprecated in June 2019 for 32-bit Windows systems. Support for the 32-bit Windows emulator continues until June 2020, including critical bug fixes, but no new features will be added. If you are using the emulator on a 32-bit Windows system, you should plan to migrate to a 64-bit Windows system.

If you are using the emulator on a 32-bit Windows system, you can use the SDK Manager to install the latest version of the emulator for 32-bit Windows.

Following are the dependencies used during implementation

Razorpay: (razorpay_flutter: ^1.2.6**)**

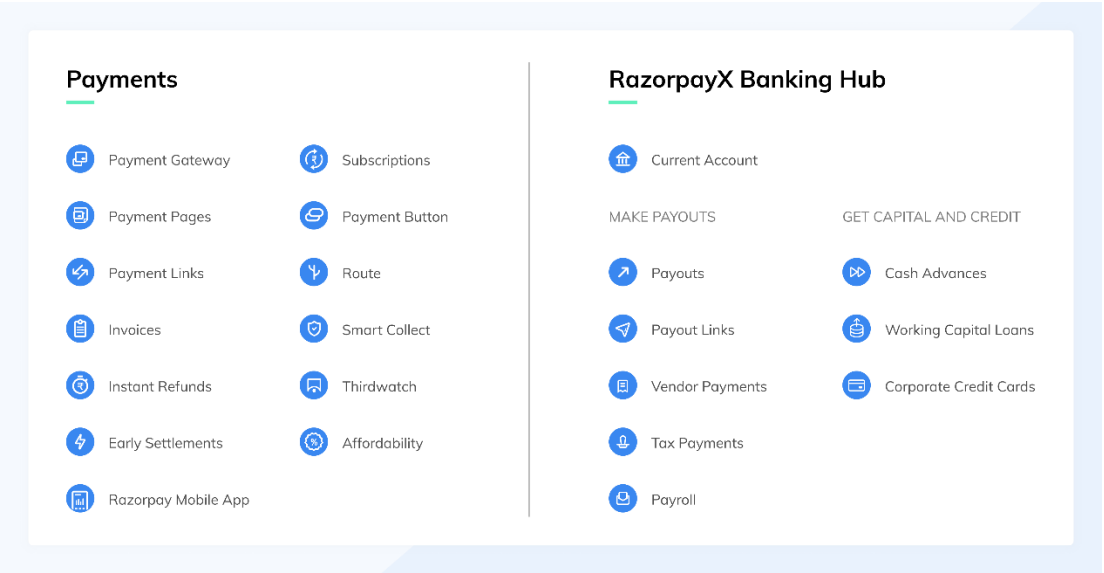


Fig 4.1 RazoprPay Solutions.

RazorPay aims to revolutionize money management for online businesses by providing clean, developer-friendly APIs and hassle-free integration. Razorpay offers a range

of solutions for all your payment and finance needs. Solutions offered by Razor Pay is as shown in above Fig 4.1.

We offer a fast, affordable and secure way for merchants, schools, ecommerce and other companies to accept and disburse payments online, own a fully functional current account and avail working capital loans.

Carousel Slider:(carousel_slider: ^2.3.1**)**

Carousel Slider is one of the most popular image slider used nowadays in most apps. These Carousel Sliders are mostly seen in various eCommerce sites such as Amazon, Flipkart, Myntra, and many more. Displaying the images in a slider gives an attractive User Experience. As these sliders are automated you can get to see various types of images and content in them.

Properties of Carousel Slider:

1.Items: In which we must declare Asset Images or Network Images that are used in our app.
2.Options: It consists of many properties such as:
3.Height: Overall height of card to display the image.
4.Autoplay: Cards automatically slides at a time.
5.Autoplaycurve: Determines the animation curve.
6.AspectRatio: Aspect Ratio is used to declare the height.
7.AutoPlayAnimationDuration: Used to declare time for automated slide.

Autocomplete TextField: (autocomplete_textfield: ^1.7.3**)**

AutoCompleteTextField supports any data type suggestions. New AutoCompleteTextField<YOURDTATYPE>().The suggestions parameter must have data that matches <YOURDATATYPE> A global key of type GlobalKey <AutoCompleteTextFieldState<T>> is required so that the clear() method can be called to clear AutoCompleteTextField.

Toast: (toast: ^0.1.5**)**

Flutter Toast is also called a **Toast Notification** message. It is a very small message which mainly popup at the bottom of the device screen. It will disappear on its own after completing the time provided by the developers. A developer mostly used the toast notification for showing feedback on the operation performed by the user.

Showing toast notification message is an essential feature in android applications. We can achieve it by using simple lines of code. In this section, we are going to learn how to show toast message in android and iOS by implementing it in Flutter. To implement toast notification, we need to import **fluttertoast** library in Flutter.

The following steps are needed to show toast notification in Flutter:

- Create a Flutter Project.
- Add the Flutter Toast Dependencies in project.
- Import the fluttertoast dart package in library.
- Implement the code for showing toast message in Flutter.

Wiredash: (wiredash: ^0.6.2**)**

Wiredash is probably the easiest, and most convenient way to capture in-app user feedback, wishes, ratings and much more. The SDK is completely written in Dart and runs on Android, iOS, Desktop and the Web. For more info, head over to wiredash.io.

Clay containers: (clay_containers: ^0.2.2**)**

Easily create and customize beautiful, modern neumorphic containers for your Flutter project. These clay containers can become the basis for your own unique neumorphic designs.

Named Arguments:

- color - This sets the base color for the clay object. Simply setting this to the background color of the parent object will give you a pretty good looking debossed neumorphic effect.

- height - This sets the height of the container.
- width - This sets the width of the container.
- parentColor - This tells the widget to use a different color for the outside emboss/deboss effect, despite whatever is set in the color field.
- surfaceColor - This tells the widget to use a different color for the inside of the container, despite whatever is set in the color field.

Lottie: (lottie: ^0.7.0+1**)**

Lottie is a mobile library for Android and iOS that parses Adobe After Effects animations exported as json with Bodymovin and renders them natively on mobile!.This repository is an unofficial conversion of the Lottie-android library in pure Dart.

It works on Android, iOS, macOS, linux, windows and web.Lottie files are a simple Flutter app with only one page. The screen has an app bar and Center widget to place the animation in the middle of the screen. The Lottie.network() takes the URL of the JSON animation file and renders the animation.

- The Lottie.network()'s reverse property is used to reverse the motion in the animation.
- The Lottie.network()'s repeat property is used to keep repeating the animation.

Firebase Authentication:(firebase_auth: ^1.2.0**)**

Firebase Authentication provides backend services, easy-to-use SDKs, and readymade UI libraries to authenticate users to your app. It supports authentication using passwords, phone numbers, popular federated identity providers like Google, Facebook and Twitter, and more.

Firebase Authentication integrates tightly with other Firebase services, and it leverages industry standards like OAuth 2.0 and OpenID Connect, so it can be easily integrated with your custom backend.

Email and password Authentication: Authenticate users with their email addresses and passwords. The Firebase Authentication SDK provides methods to create and manage users that use their email addresses and passwords to sign in. Firebase Authentication also handles sending password reset emails.

Firebase core :(firebase_core: ^1.2.1**)**

A Flutter plugin to use the Firebase Core API, which enables connecting to multiple Firebase apps.

The firebase_core plugin is responsible for connecting your Flutter app to your Firebase project. The plugin must be installed and initialized before the usage of any other FlutterFire plugins. It provides basic functionality such as:

- Initializing FlutterFire.
- Creating Secondary Firebase App Instances.

Cloud Firestore: (cloud_firestore: ^1.0.7**)**

Cloud Firestore is a flexible, scalable database for mobile, web, and server development from Firebase and Google Cloud. Like Firebase Realtime Database, it keeps your data in sync across client apps through realtime listeners and offers offline support for mobile and web so you can build responsive apps that work regardless of network latency or Internet connectivity. Cloud firestore also offers seamless integration with other Firebase and Google Cloud products, including Cloud Functions.

Cloud Firestore stores data in Documents, which are stored in Collections. It creates collections and documents implicitly the first time you add data to the document. You do not need to explicitly create collections or documents.If you're using the Web, Android, or iOS SDK, use Firebase Authentication and Cloud Firestore Security Rules to secure your data in Cloud Firestore.

URL Launcher: (url_launcher: ^6.0.5**)**

A Flutter plugin for launching a URL. Supports iOS, Android, web, Windows, macOS, and Linux.The launch method takes a string argument containing a URL. This URL can be formatted using several different URL schemes. The supported URL schemes depend on the underlying platform and installed apps. URLs must be properly encoded, especially when including spaces or other special characters. This can be done using the Uri class.

A particular mobile device may not be able to receive all supported URL schemes. For example, a tablet may not have a cellular radio and thus no support for launching a URL using the sms scheme, or a device may not have an email app and thus no support for launching a URL using the email scheme.

We recommend checking which URL schemes are supported using the canLaunch method prior to calling launch. If the launch method returns false, as a best practice we suggest adjusting the application UI so that the unsupported URL is never triggered; for example, if the email schemeis not supported, a UI button that would have sent email can be changed to redirect the user to a web page using a URL following the http scheme.

METHODOLOGY:

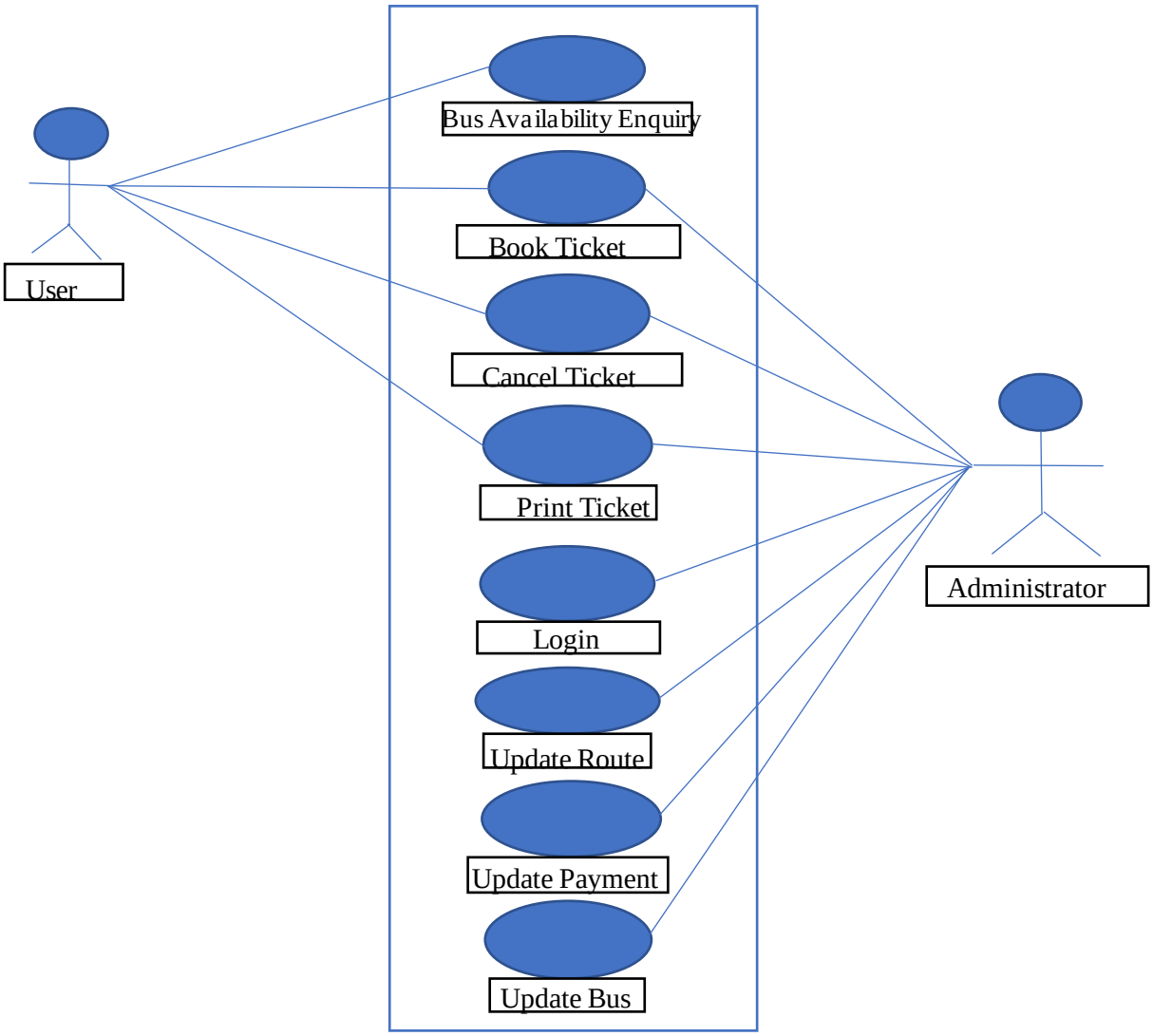


Fig 4.2 Use case diagram for users and admin.

User End: It contains user end where users can access bus availability for their desired places, booking tickets, take a copy of ticket confirmation as shown in Fig 4.2 and many other features are available.

Admin End: In the administration end, admin can access user end as well as login details, Updating the route, Updating the payment in Razor Pay, Updating the buses as shown in Fig 4.2.

CHAPTER-5

RESULTS

Fig 5.1 Splash Screen.

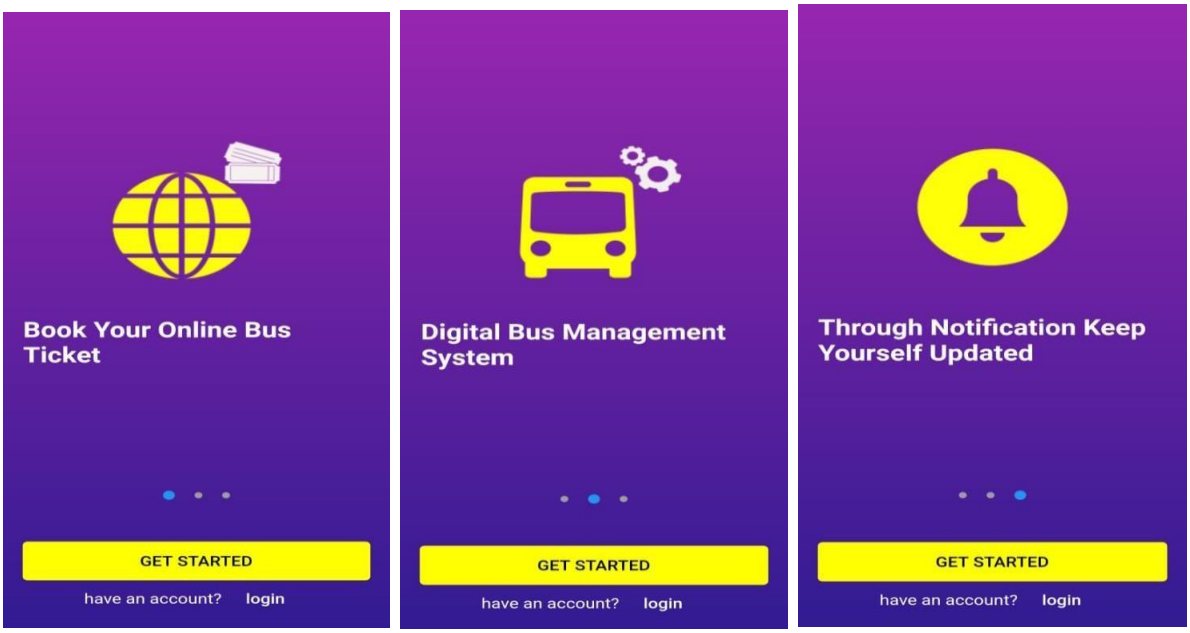


Fig 5.2 Carousel Sliding Windows.

When user opens the app splash screen is displayed as shown in Fig 5.1. The user gets overview of our app from Fig 5.2 i.e., Carousel Sliding Windows. The screen displayes users can Book Tickets online through Digital Bus Management System and get updated through Notifications. User can click on “Get Started” to Sign Up for our app.

Fig 5.3 Signup Screens.

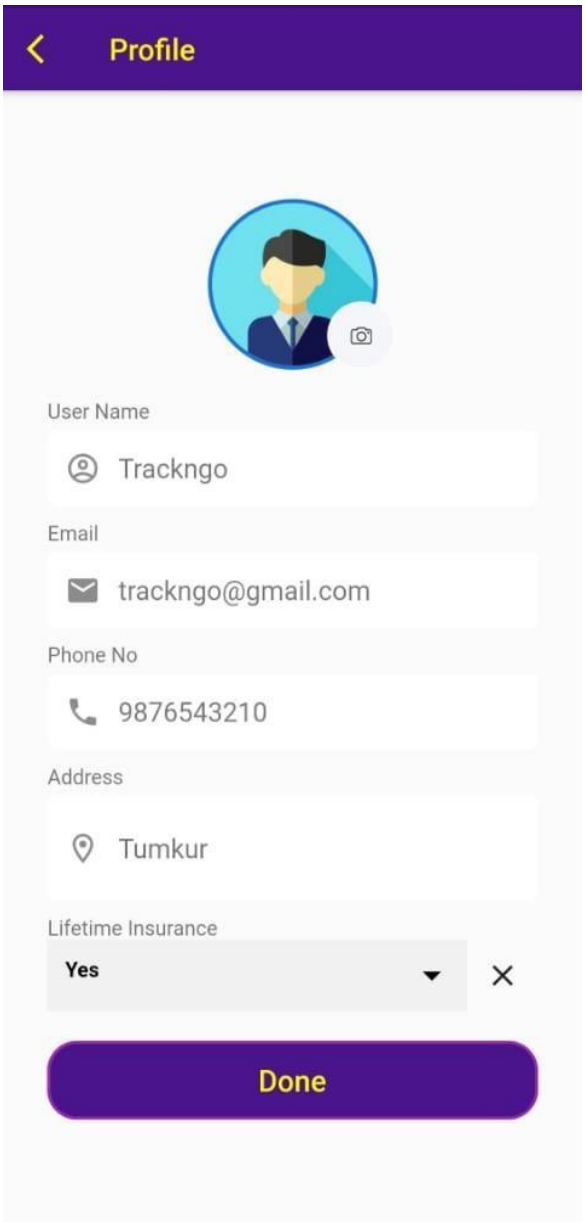


Fig 5.4 User Details Form.

User can sign up themselves by using their mail id and by setting password to the app as shown in Fig 5.3. User can enter their details like Username, Email, Phone No., Address and Life Insurance (if any) as shown in Fig 5.4.

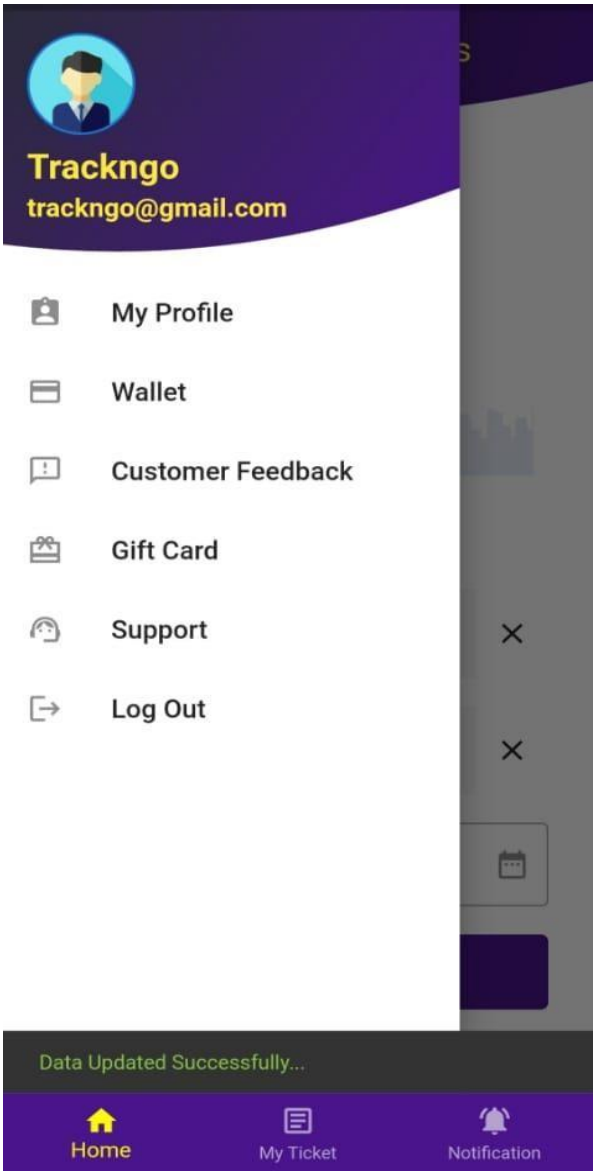


Fig 5.5 Drawer Screen.

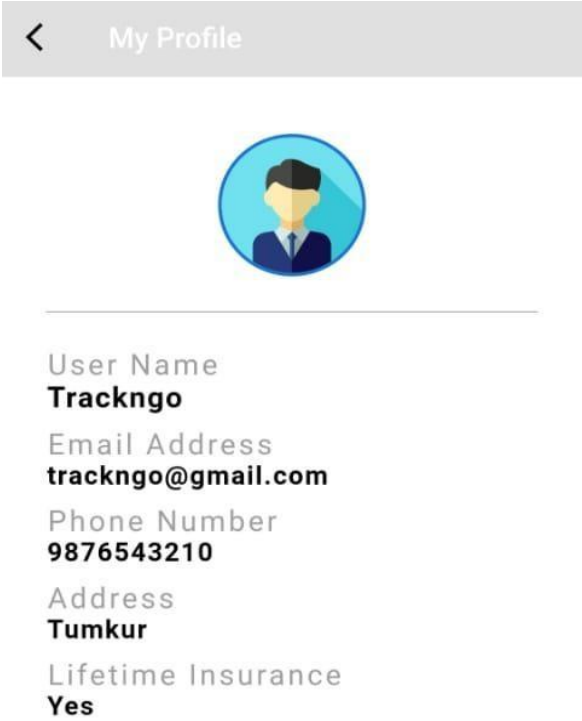


Fig 5.6 Profile Details.

This is a drawer screen which contains users Profile, Wallet, Customer Feedback, Gift Card, Support options as shown in Fig 5.5. It helps the user to have quick access to these options continuously. Fig 5.6. contains user profile with details like Username, Email Address, Phone Number, Address and Life Insurance which is entered by the user during Sign up. The details will be displayed.

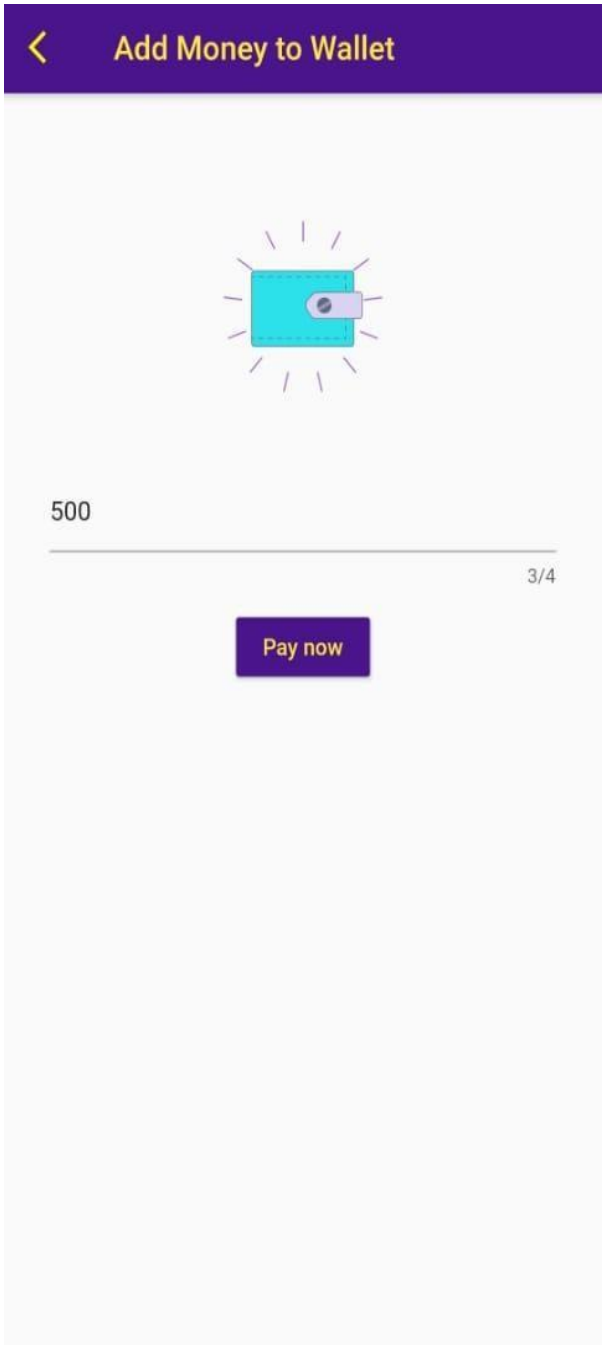


Fig 5.7 Add money to Wallet.

Fig 5.8 Gift Cards.

User can add money to any wallet like PayTM, PayZapp, etc as shown in Fig 5.7. The user can add desired amount to wallet and can use them later during ticket booking. Gift cards can be used by the user to get discounts during their payments. The user can use this utility as shown in above Fig 5.8.

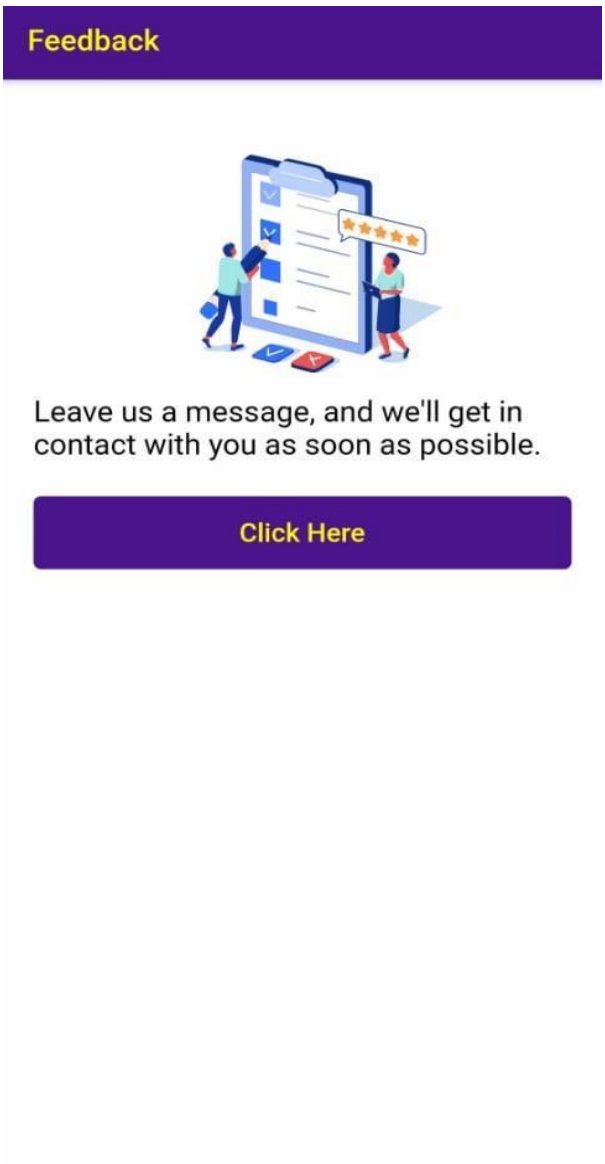


Fig 5.9 Feedback.

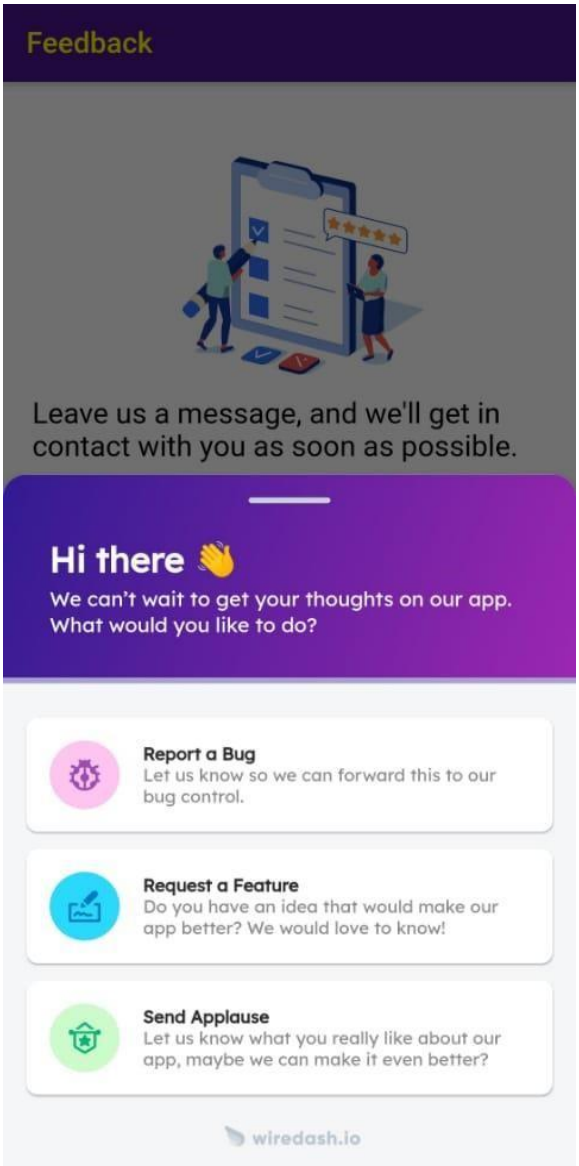


Fig 5.10 Features provided for Feedback

User can ask for the query before journey and can also give their opinion about the journey regarding the app usage by using “Click Here” Button as shown in Fig 5.9. User can give the feedback by using the options like Report a Bug, request a Feature, Send Applause as shown in Fig 5.10 and can give feedback by taking the screenshot from the app.

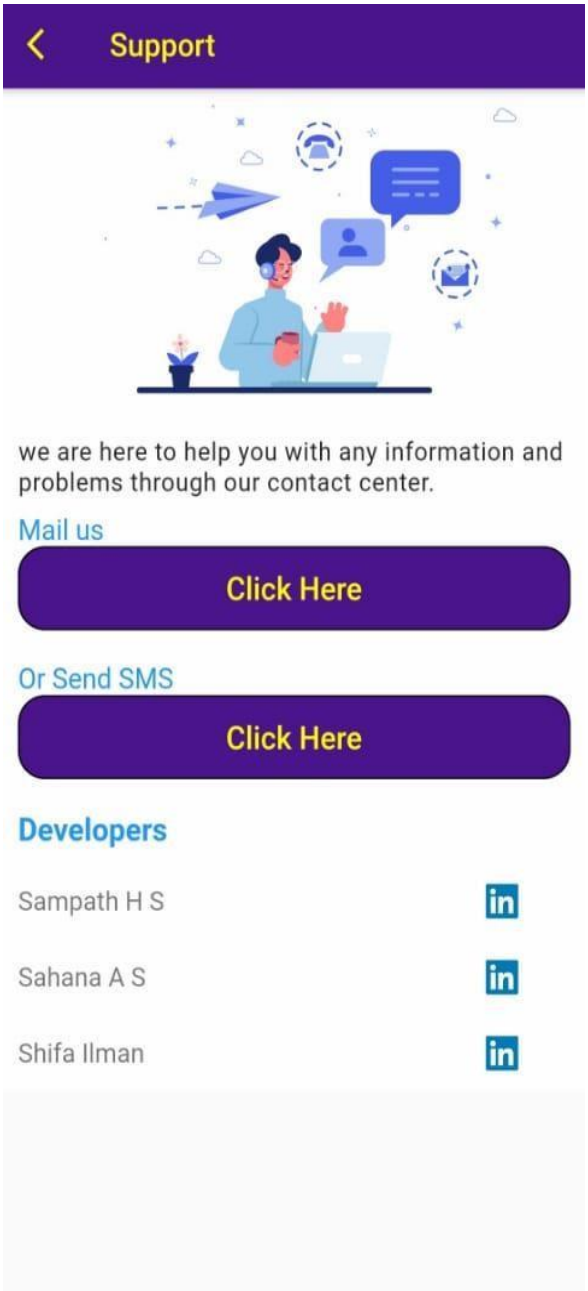


Fig 5.11 Support Screen.

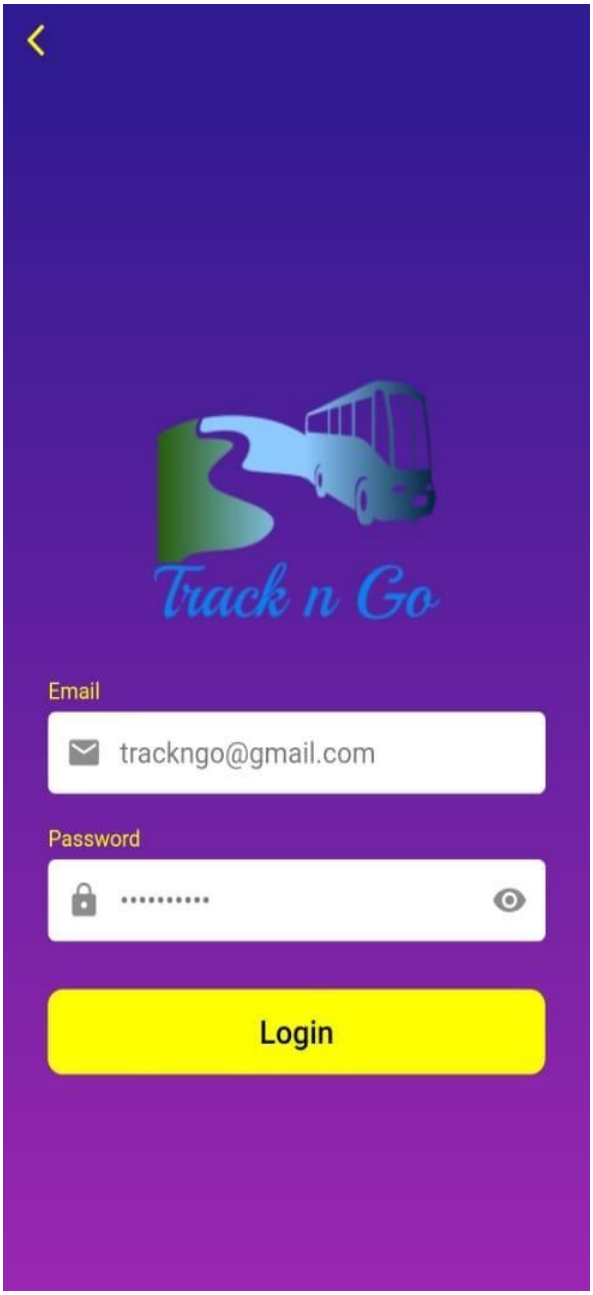


Fig 5.12 Login Screen.

User can mail or send an SMS to the developers for any query by regarding the app using respective “Click Here” Button as shown in Fig 5.11. Users can login using their Sign-Up Credentials as shown in Fig 5.12.

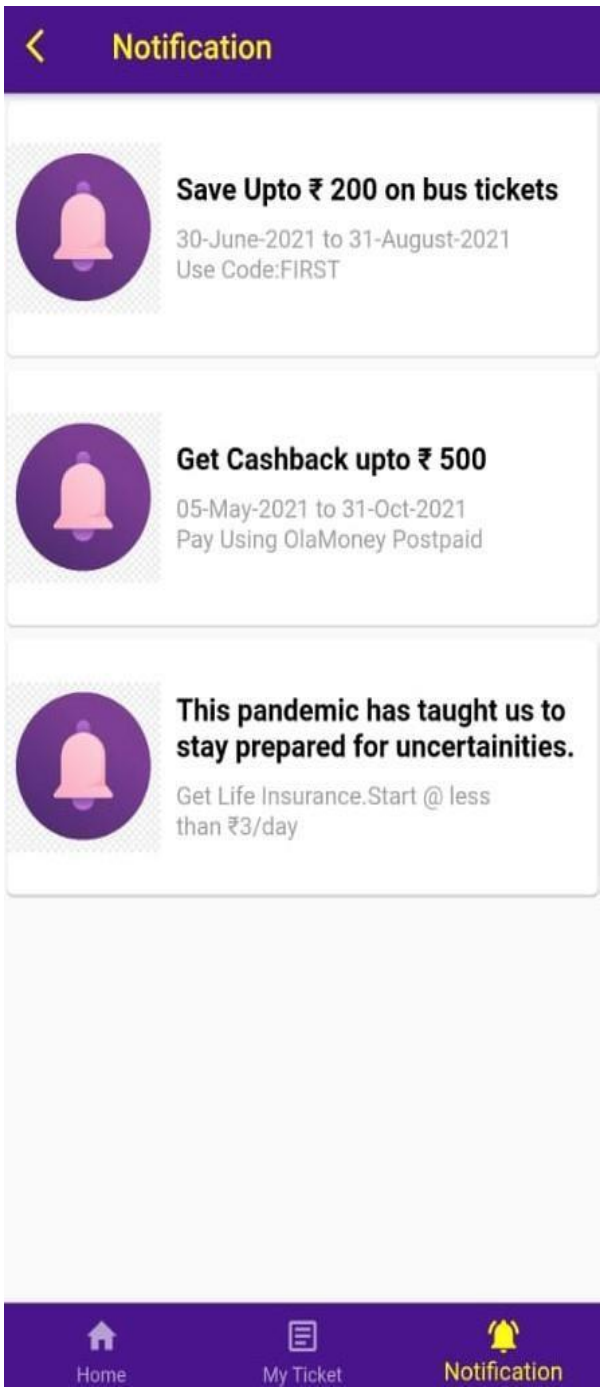


Fig 5.13 Notification Screen.

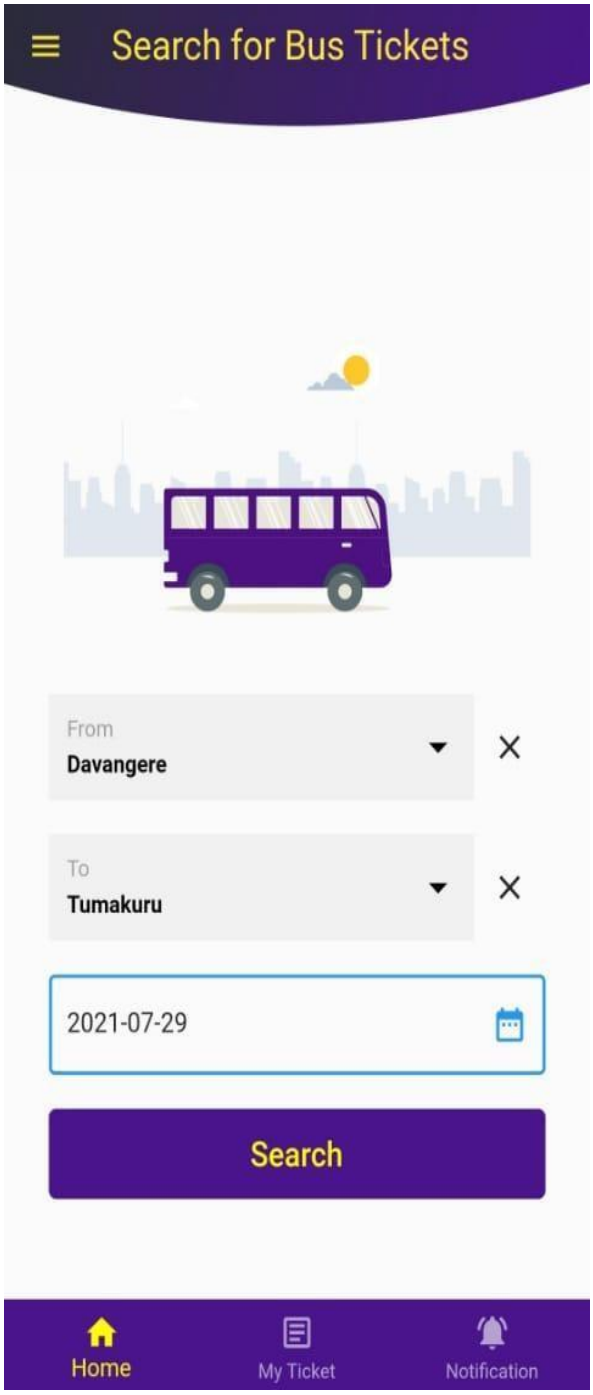


Fig 5.14 Search Screen.

Users can use this screen to see notification to get discounts on their booking by using the above coupons as shown in Fig 5.13. Users can select the Source, Destination and Date of travel for their journey in the search screen as shown in Fig 5.14.

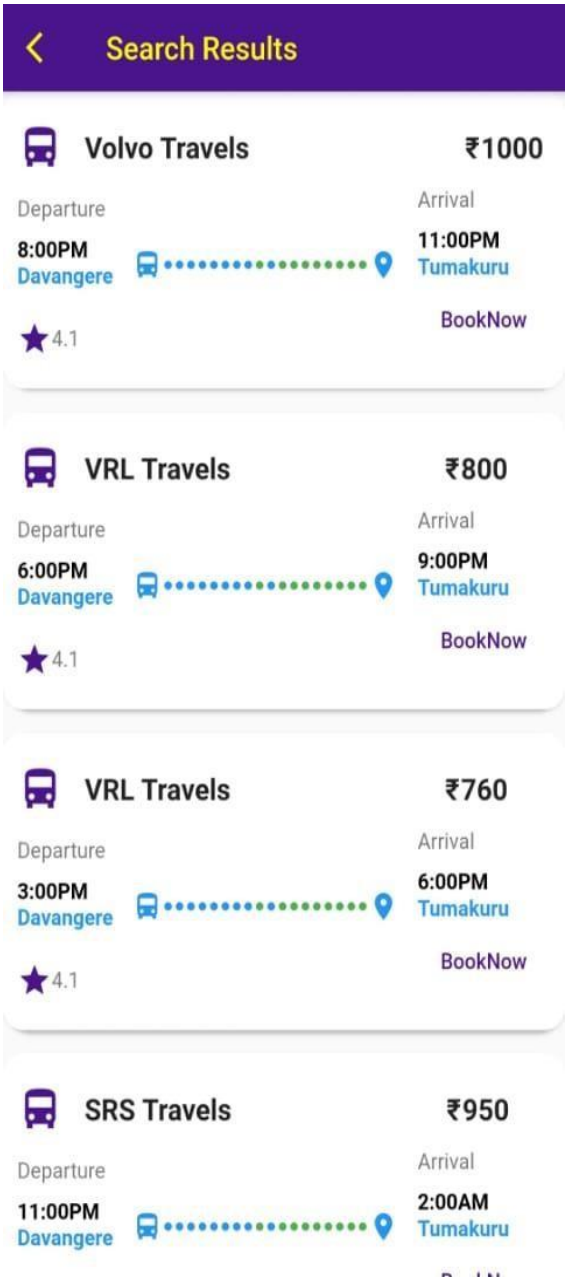


Fig 5.15 Search Results.

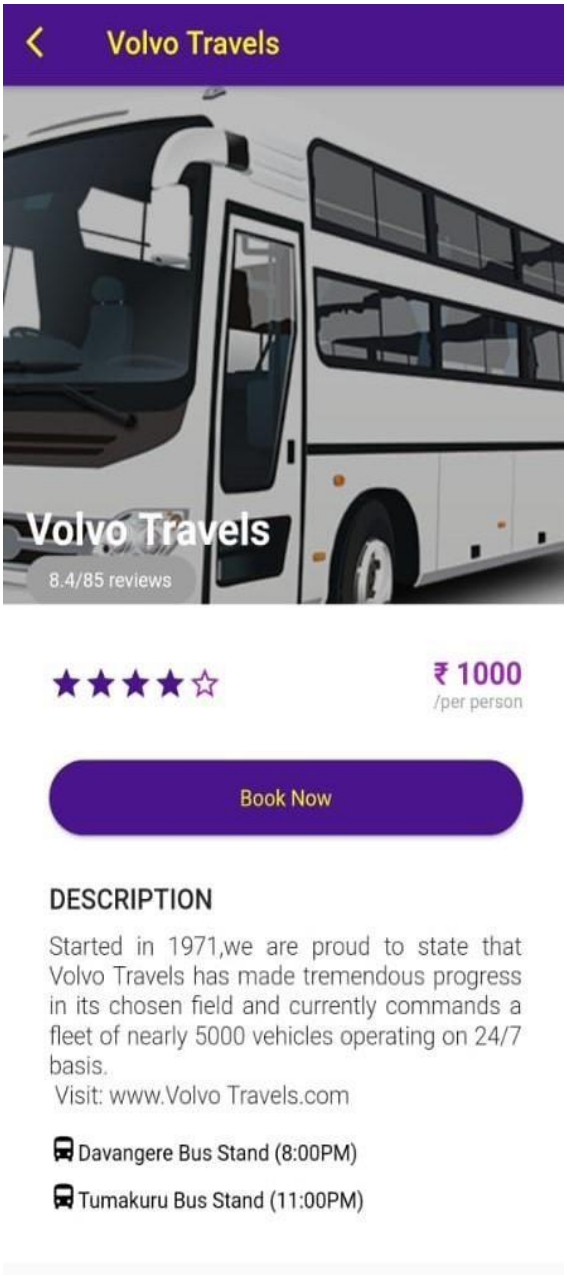


Fig 5.16 Details of selected bus

When Users selects the source and destination for their journey the respective travels name with price, time available for their journey will be displayed as shown in Fig 5.15. When users select the desired travels name its details will be displayed as shown in Fig 5.16. It also contains boarding and dropping points. Users can select “Book Now” to proceed.

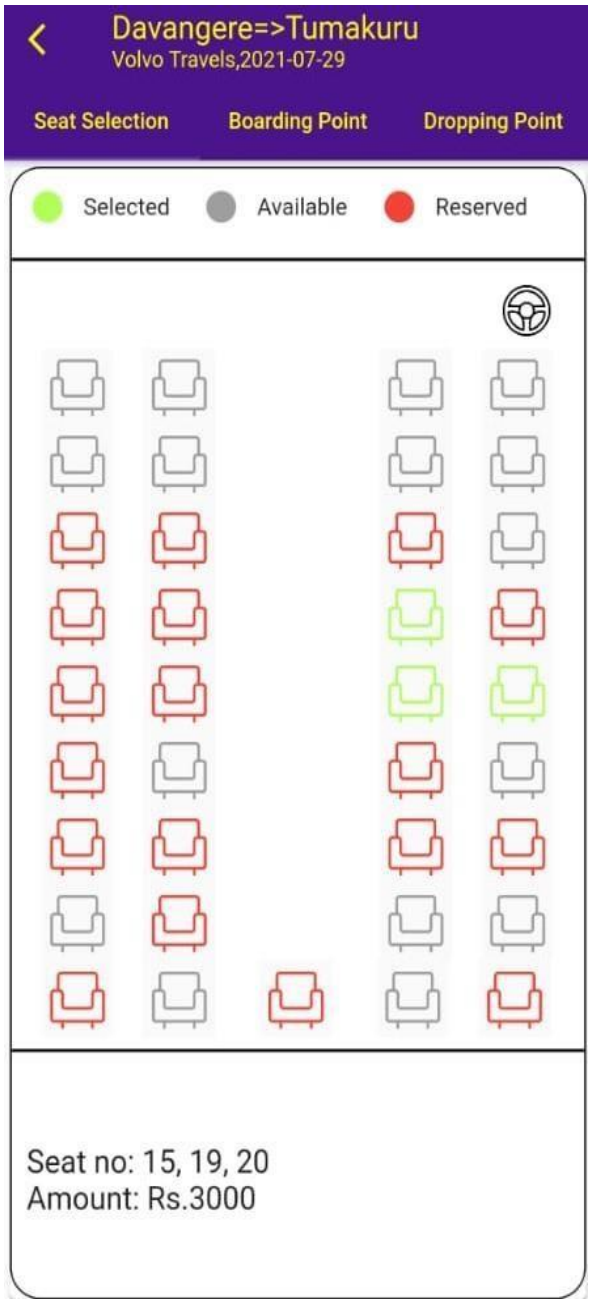


Fig 5.17 Bus Seats Selection

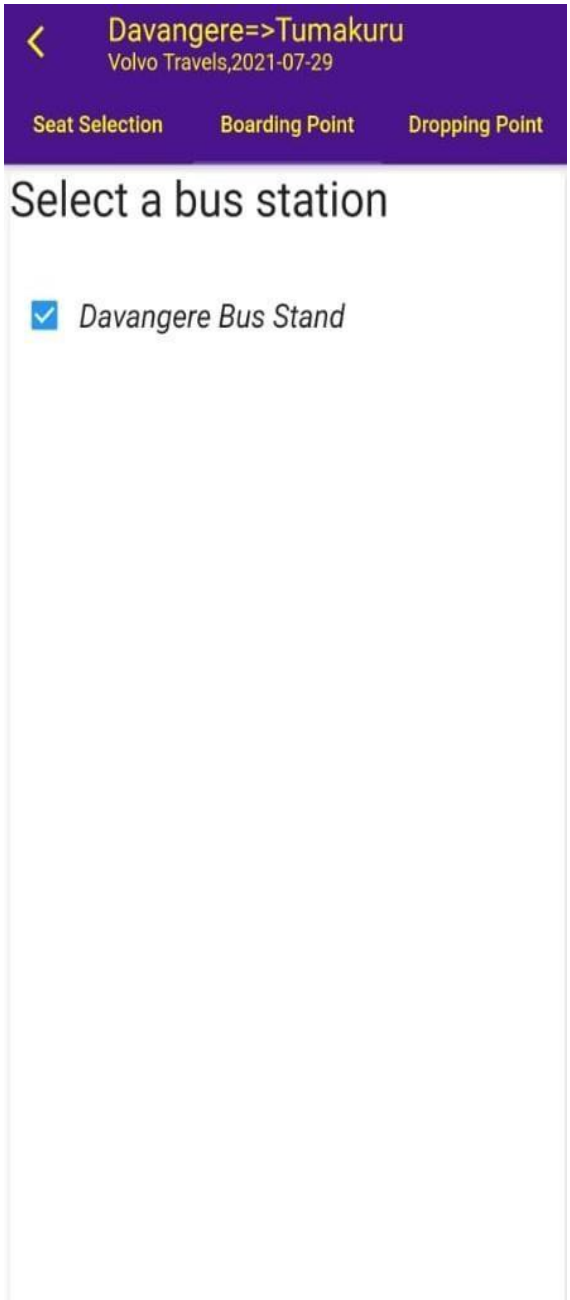
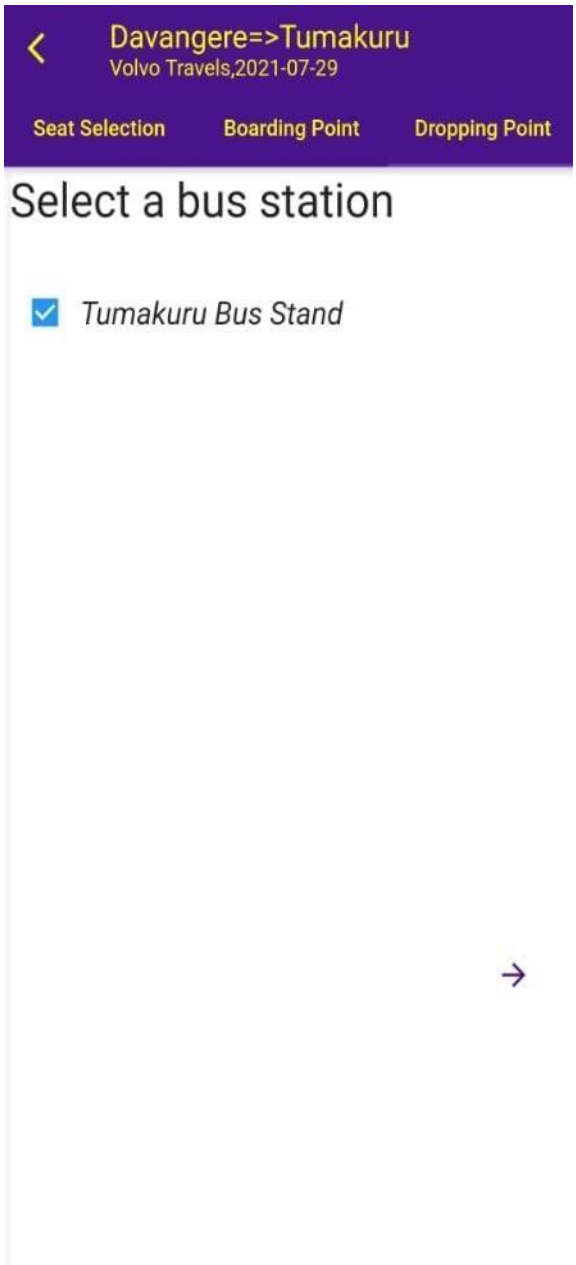


Fig 5.18 Boarding Point

After selecting "Book Now", the seats will be displayed as shown in Fig 5.17. The seats which are in red colour indicate that the seat is already booked, green indicates the selected seats and grey indicates the available seats. After the seats are selected the Boarding Point of the source is displayed as shown in Fig 5.18.



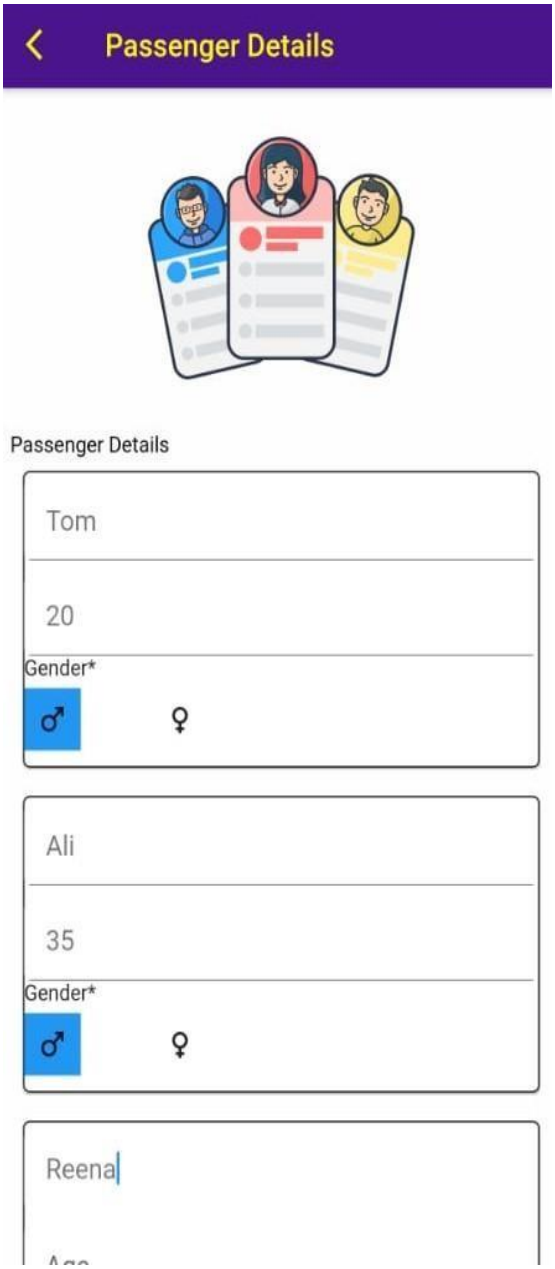


Fig 5.19 Dropping Point

Fig 5.20 Passenger Details

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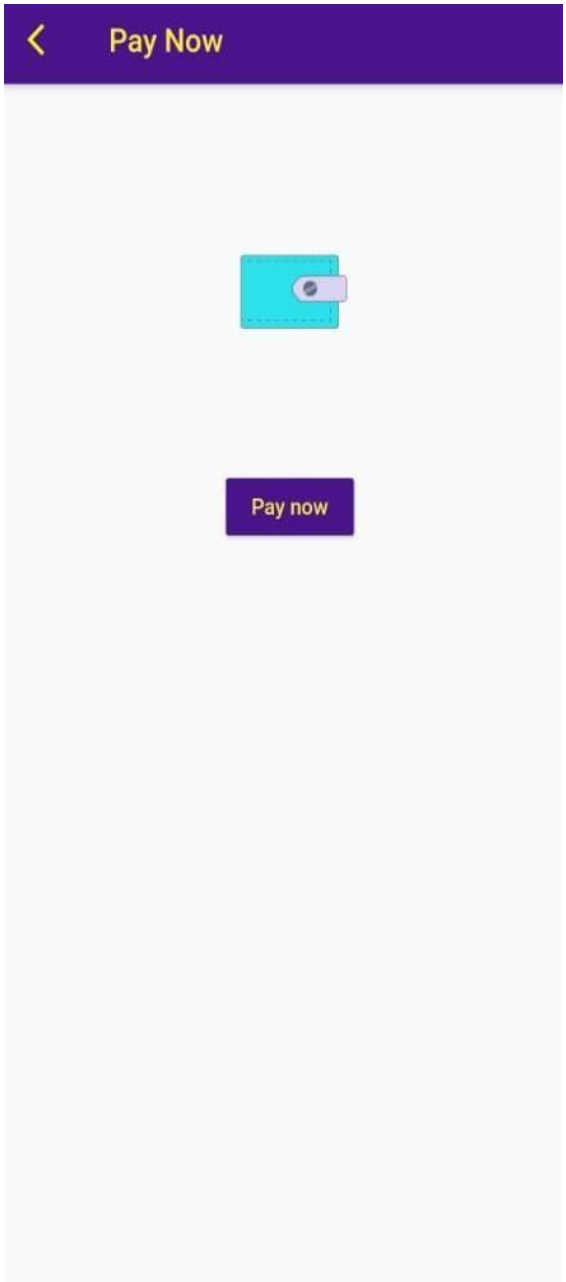
After the seats are selected the Dropping Point of the destination is displayed as shown in Fig 5.19. Passengers have to enter details like name, age and gender as shown in Fig 5.20 and should accept terms and conditions and click on “Proceed to Payment”.



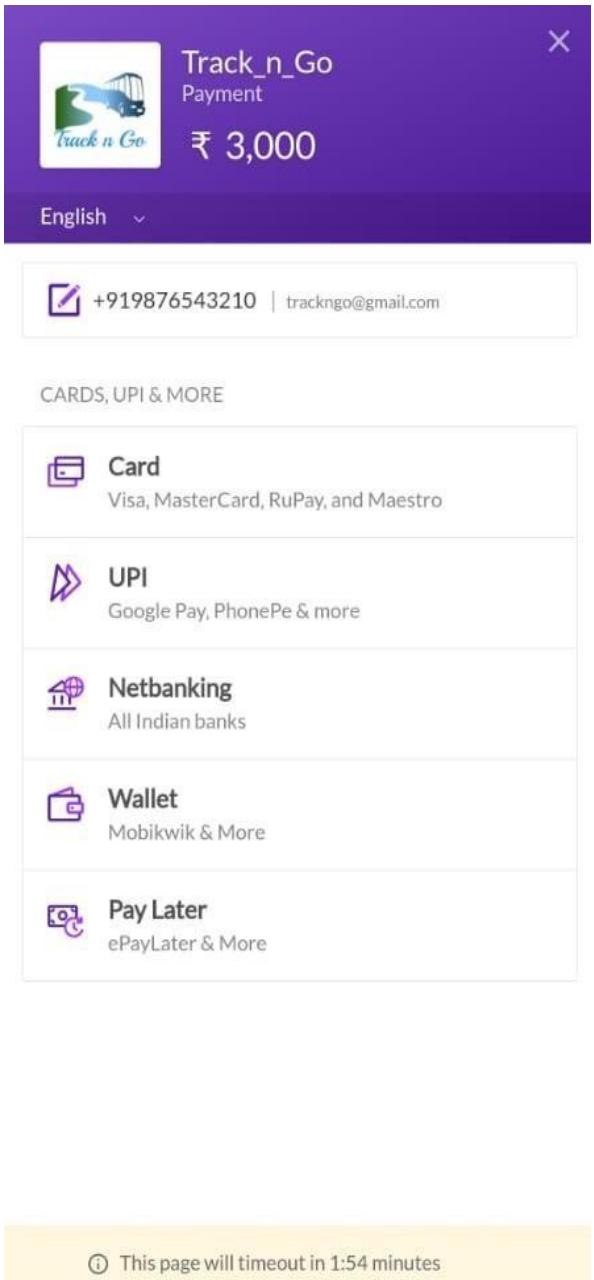


Fig 5.21 Pay Now Screen.

Fig 5.22 Payment Options.

After entering the passenger details user can click on “Pay Now” button as shown in Fig 5.21 to proceed to payment. In the payment screen user can select mode of payment like cash, UPI, Net banking, Wallet or Pay Later as shown in Fig 5.22 and can proceed.

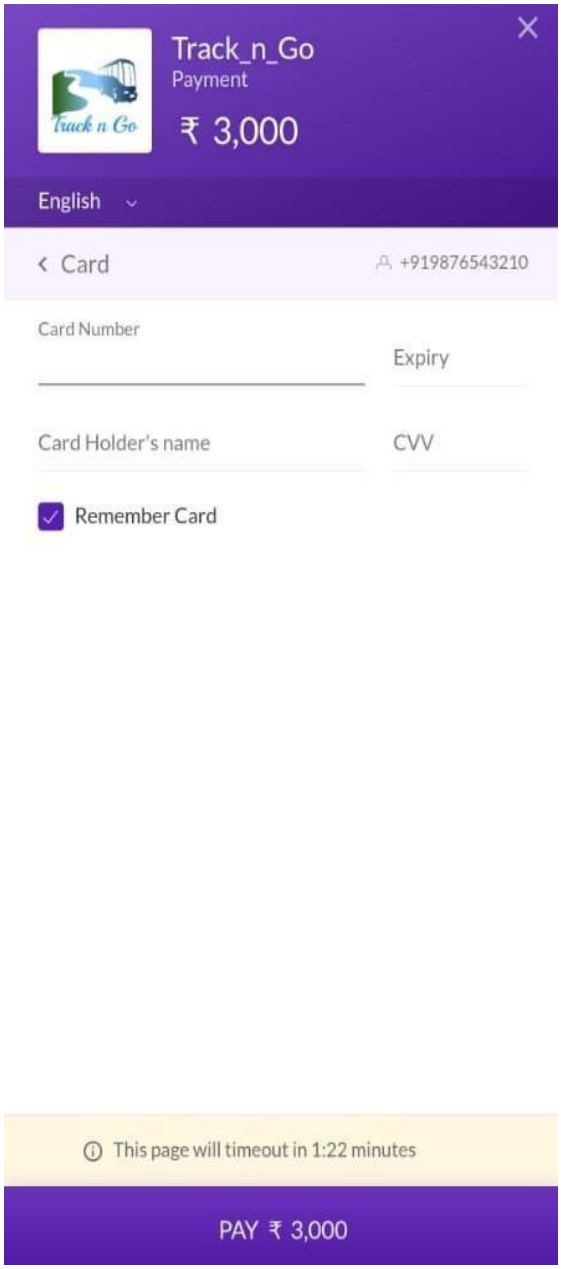


Fig 5.23 Payment Details are entered.

Fig 5.24 OTP is entered.

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User can enter details of their selected payment methods (here card details) as shown in the above Fig 5.23 and can proceed. User can enter the OTP and can complete the payment as shown Fig 5.24.

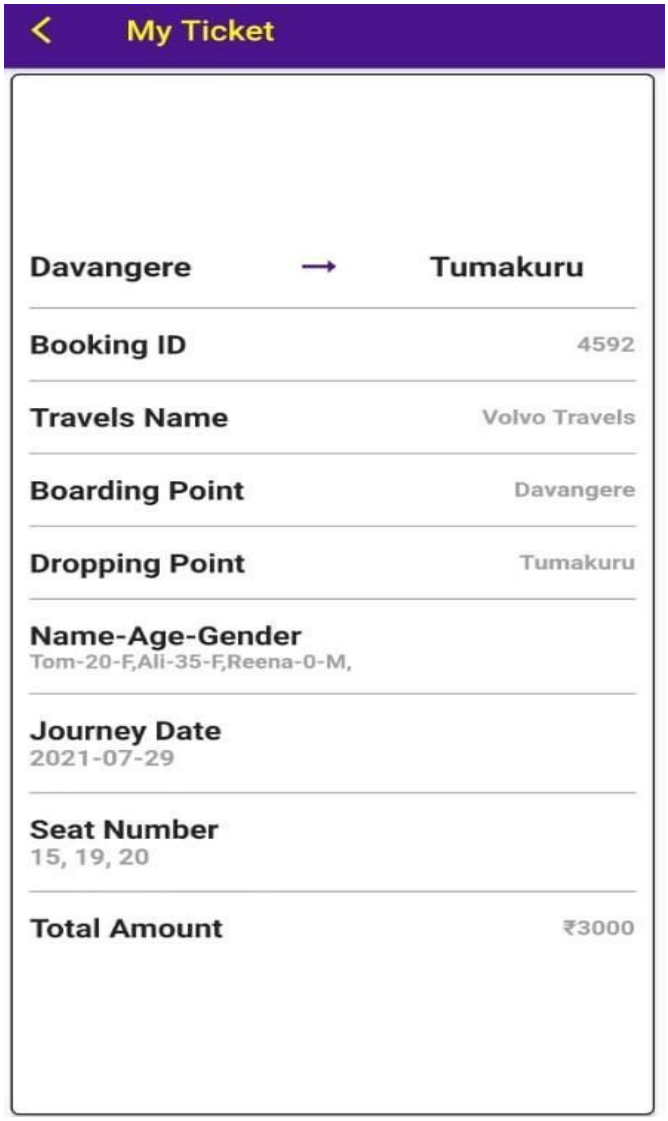

Fig 5.25 Ticket Confirmation.

After the completion of payment, ticket confirmation screen will be displayed as shown in Fig 5.25. This includes Source and destination of the journey, Booking ID, Travel's Name, Boarding Point, Dropping Point, Name-Age-Gender, Journey Date, selected Seat Numbers and Total amount.

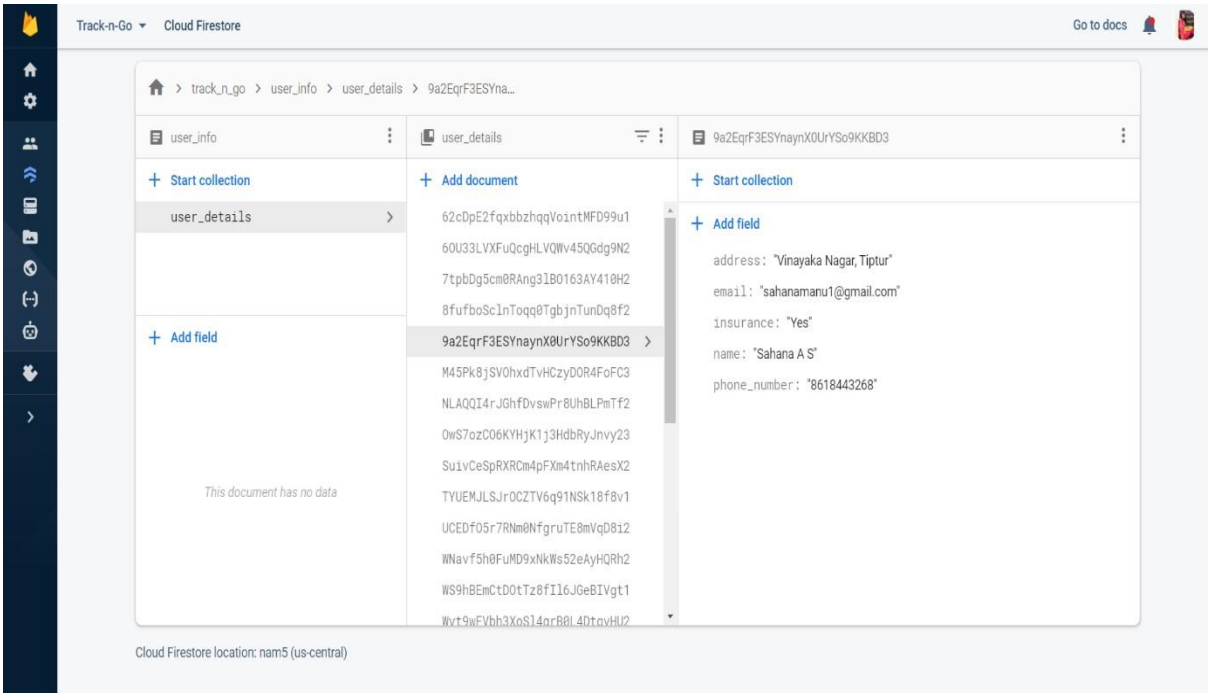

Fig 5.26 User registration details.

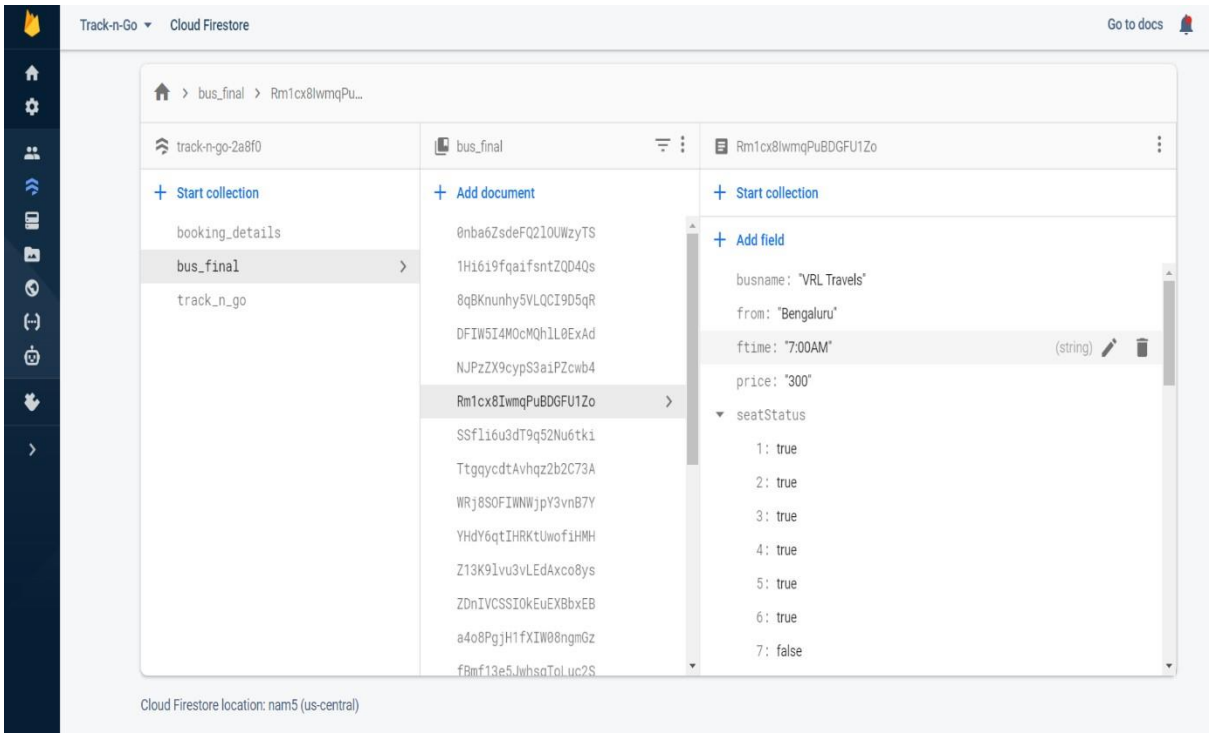

Fig 5.27 Updation bus details.

The Fig 5.26 indicates that when a user registers for our app, user details will be updated here. The following data such as Travels Name, From, To, Price, Arrival and Departure time, Seat status as shown in Fig 5.27 is retrieved from cloud firestore during search.

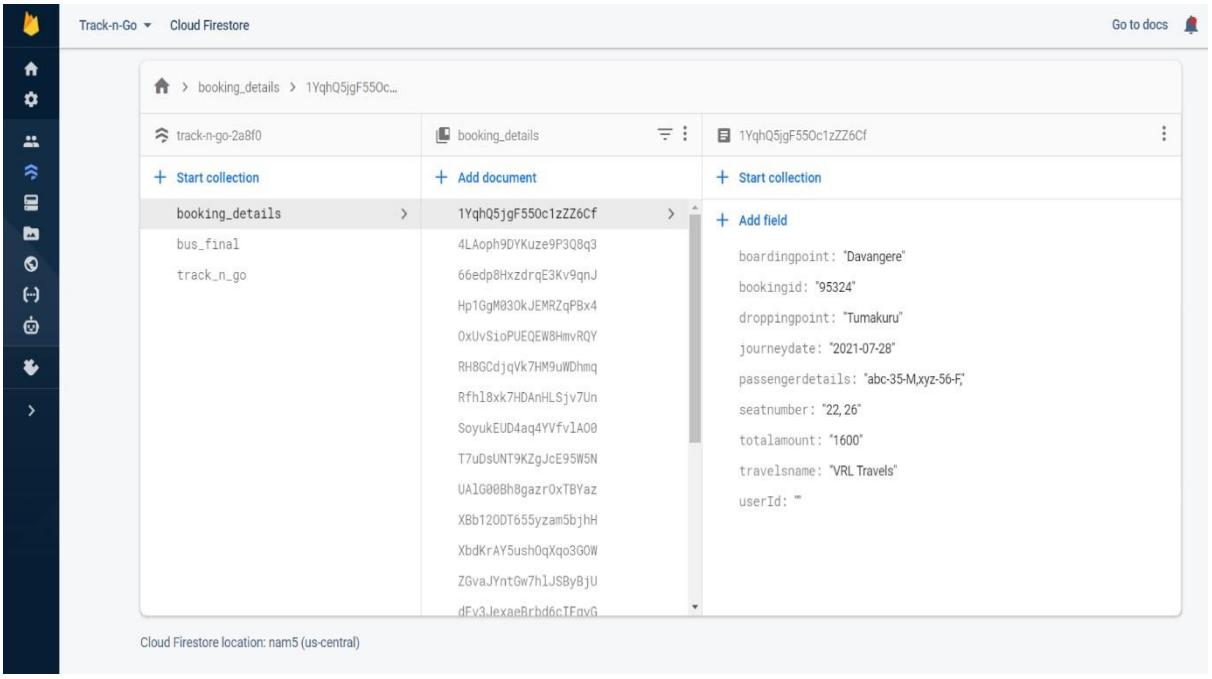

Fig 5.28 Passenger's Personal Details.

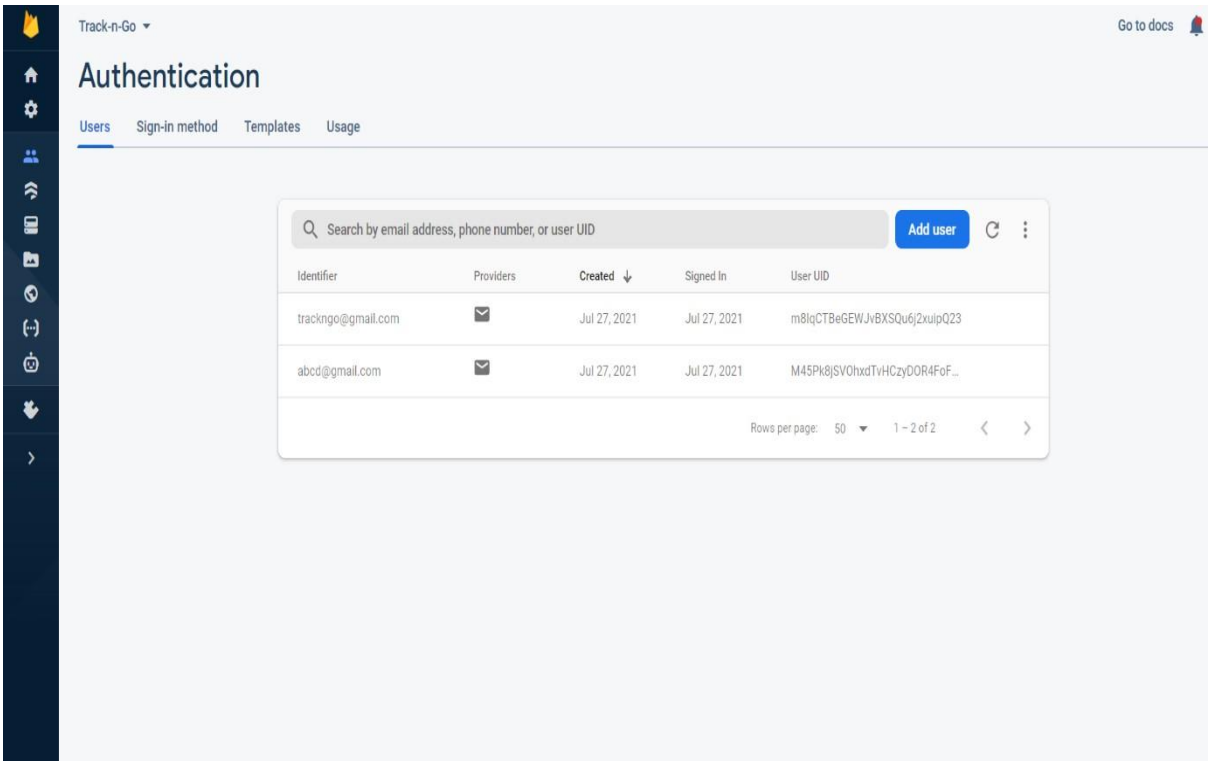

Fig 5.29 Users Registered Credentials.

Passenger details will be stored and accessed from this Collection as in Fig 5.28 which is used during ticket confirmation. Users Registered Credentials will be stored as shown in Fig 5.29 and can be accessed from here during Login.

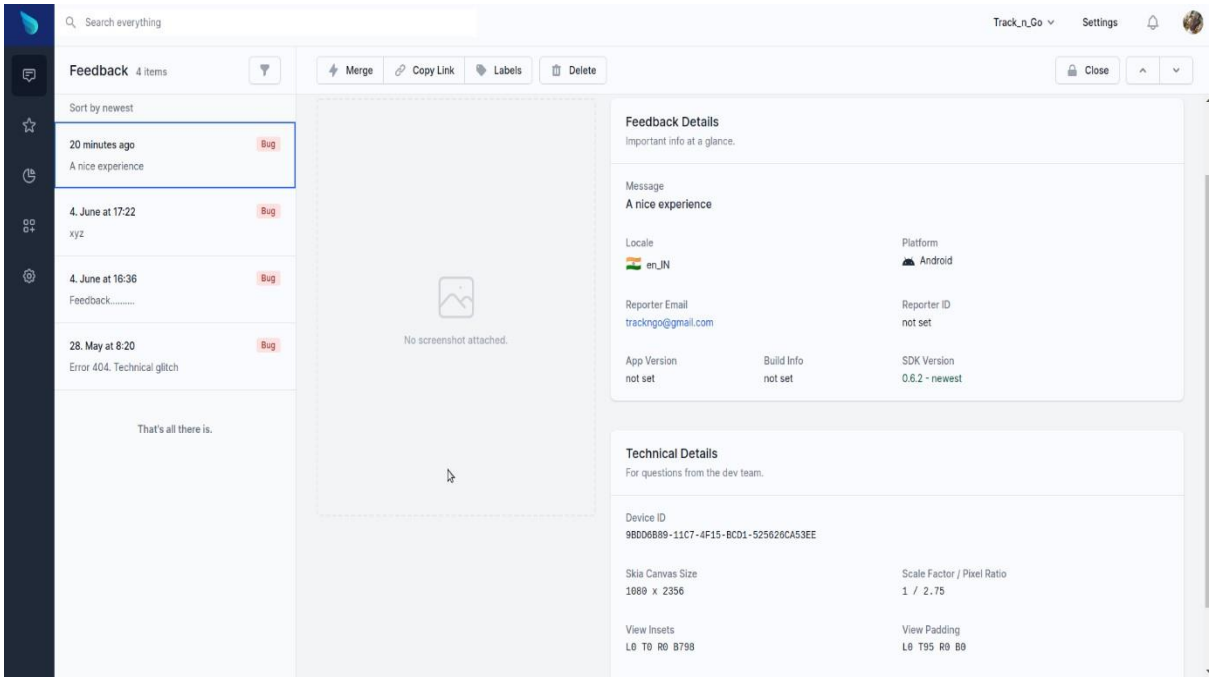

Fig 5.30 User's feedback.

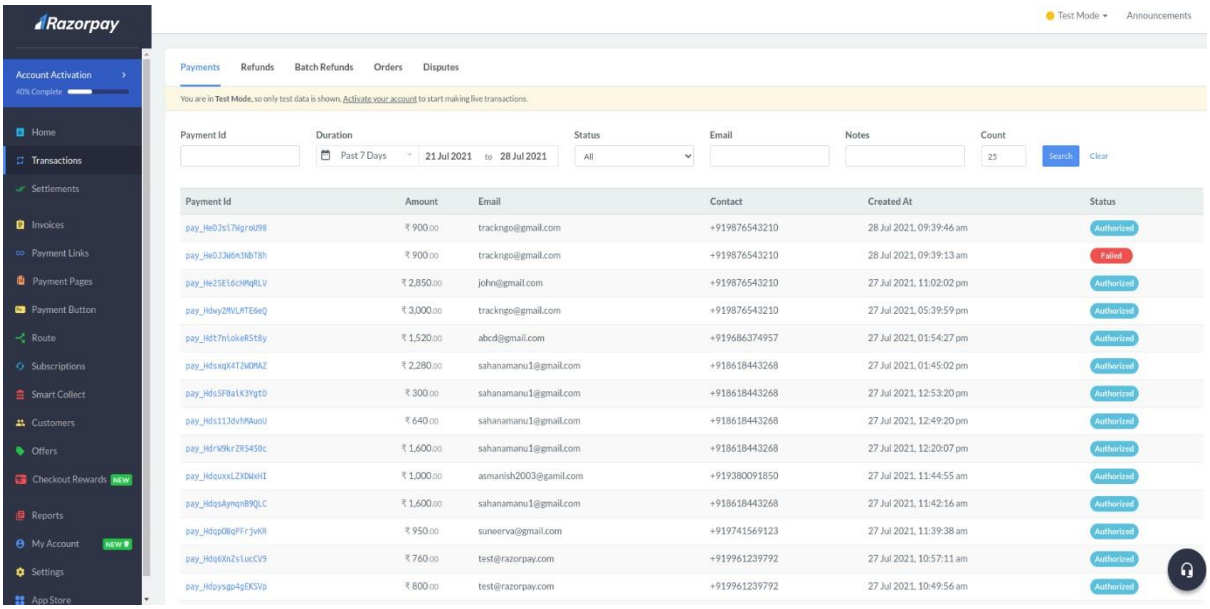

Fig 5.31 Payment Details are stored in Razorpay.

User's feedback will be stored & it can be screenshotted or mailed. It will be displayed at the administration end as in Fig 5.30. Payment Details are stored in Razorpay. Whenever a payment is completed, the developers can see the payment details as shown in Fig 5.31.

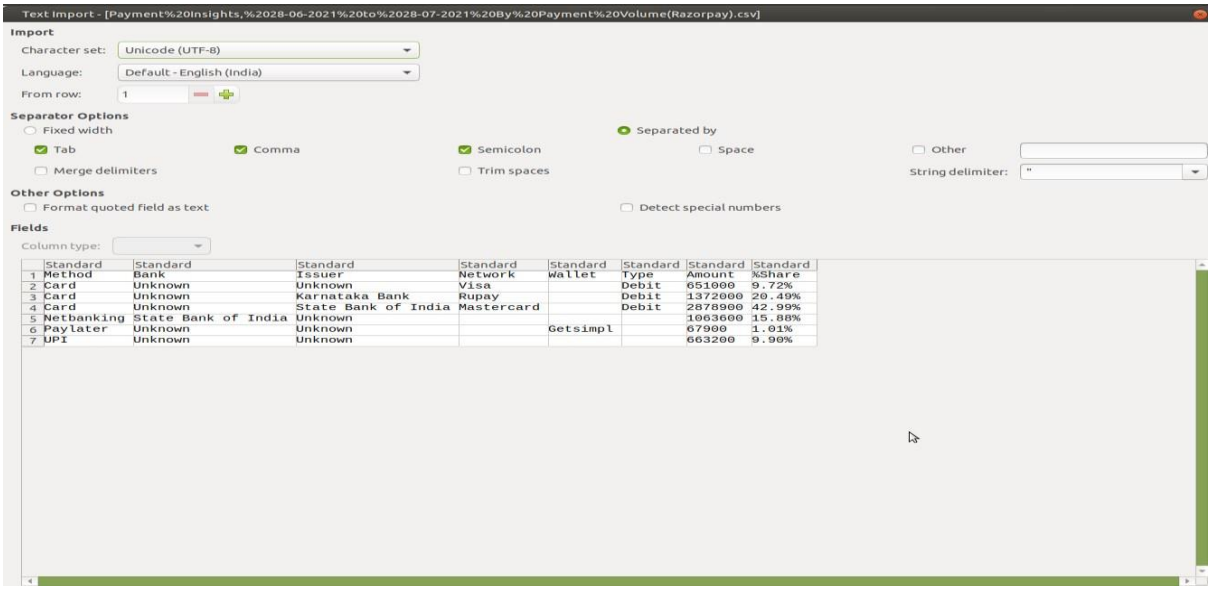

Fig 5.32 Users Payment modes are updated.

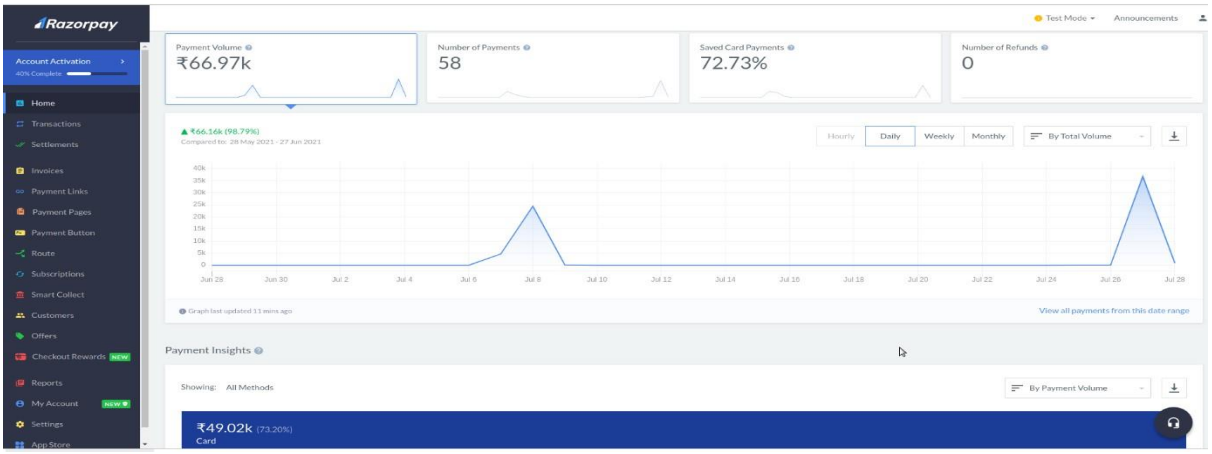

Fig 5.33 Statistics related to payment.

During payment the mode of payment selected by the user can be checked by the developers as shown in Fig 5.32. This can also be used during Refund Process. Developers can see Statistics related to payment from the Fig 5.33 which includes total amount which is being paid by all users, number of payments and saved card payments.

App Details:

Source Code: https://github.com/SampathHS/TracknGo

APK link: https://drive.google.com/drive/folders/1mSDloY5q0Fk530g6M-W1coYHwF-hZrA?usp=sharing

CONCLUSION

With the help of technologies like flutter, firebase, android studio and supportive hardware we have developed this customer friendly application that responds to the need of customers instantly.

It can be observed that computer applications are very important in every field of human endeavour. Here all the information about customer that made reservation can be gotten just by clicking a button with this new system, some of the difficulties encountered with the manual system are overcome. It will also reduce the workload of the staff, reduce the time used for making reservation at the bus terminal and also increase efficiency. The application also has the ability to update records in various files automatically thereby relieving the company's staff the stress of working from file security of data.

Finally, in Online Bus Booking System, we have developed a secure, user-friendly Bus Seat Booking App. This Project basically provides Bus information. First of all, in our app any user or visitors view our system and search the bus and how many seats are available in our buses. The user can also register their own seats on the bus on this app.

The user can post a feedback on different Bus Services. But users can compulsorily be registered first in the system. Search Bus category-wise. This system has an automated process to alert employees as well as customers who have booked their seats with the system. This is a cloud-based online system. Therefore, to access this system you only need a web app, internet connection.

This is the 24X7 available system. We observed the working of the Bus booking system and after going through it, we get to know that there are many operations, which they have to do manually. It takes a lot of time and causing many errors while data entry. Due to this, sometimes a lot of problems occur, and they were facing many disputes with customers.

To solve the above problem, and further maintaining records of passenger details, seat availability, price per seat, bill generation, and other things, we are developed a computerized booking system. By using this software, we can book tickets via internet. The customer can check the availability of buses and book selective seats.

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